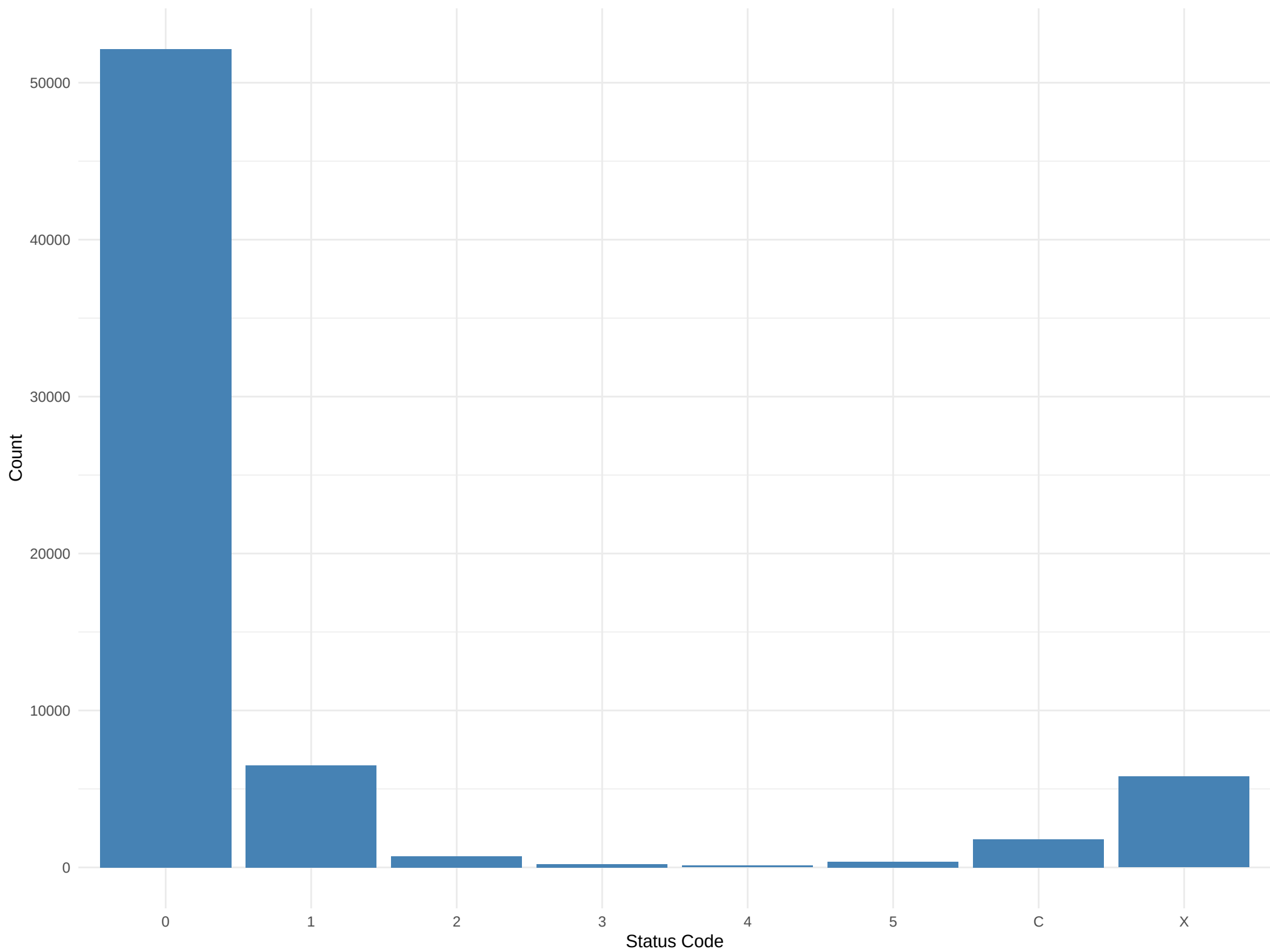
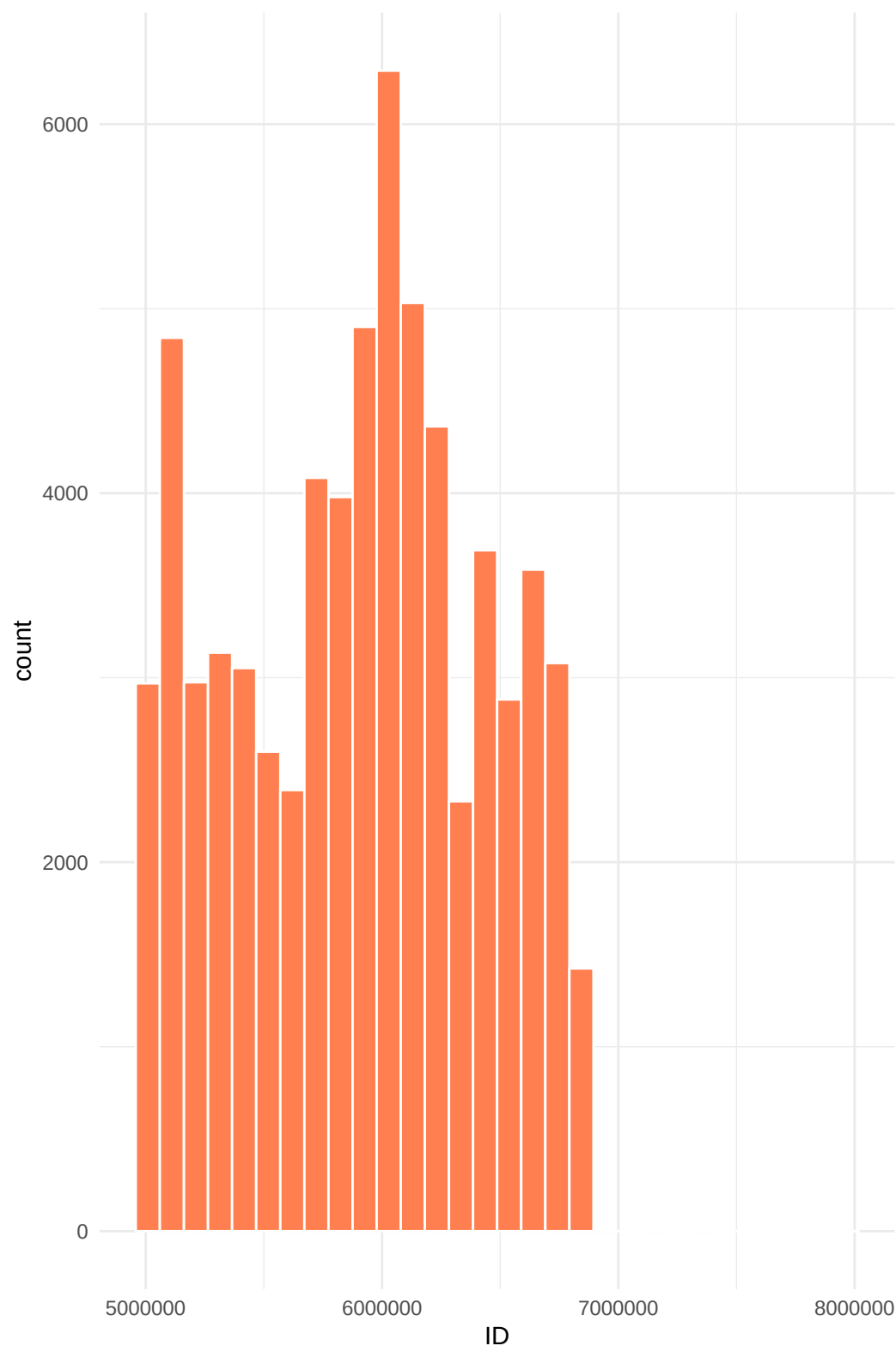


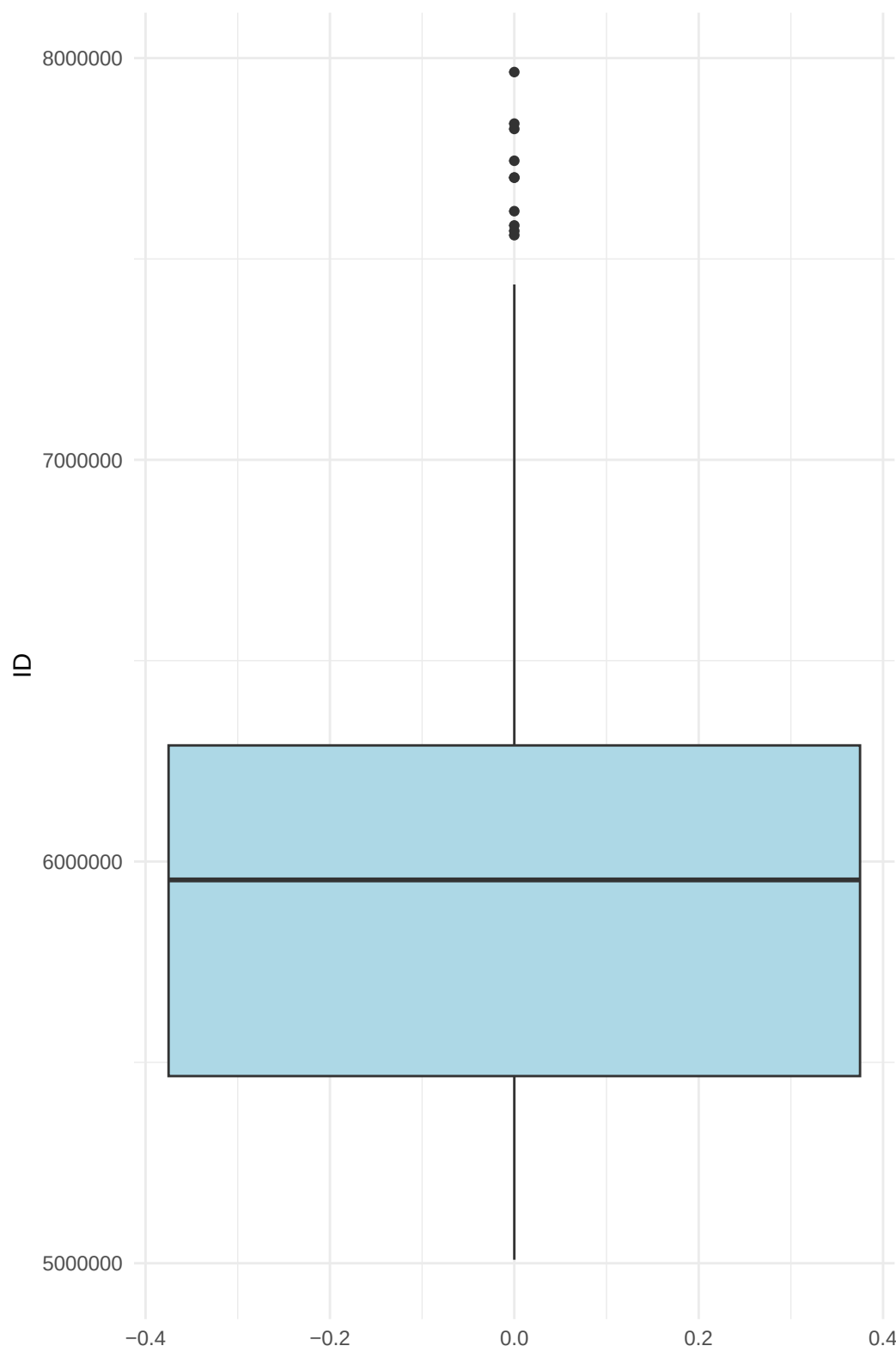
Target Distribution (Class Imbalance Check)



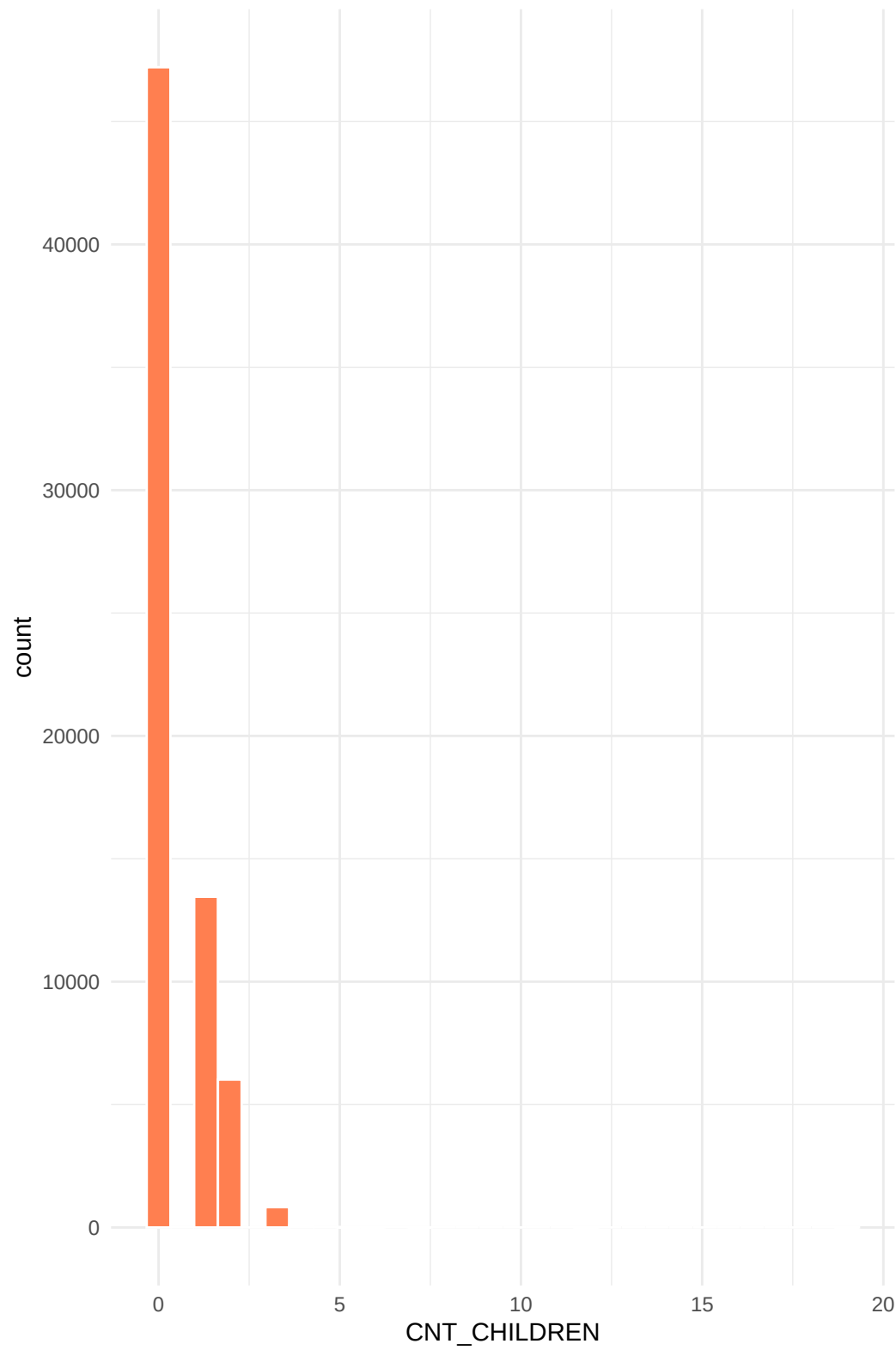
Distribution: ID



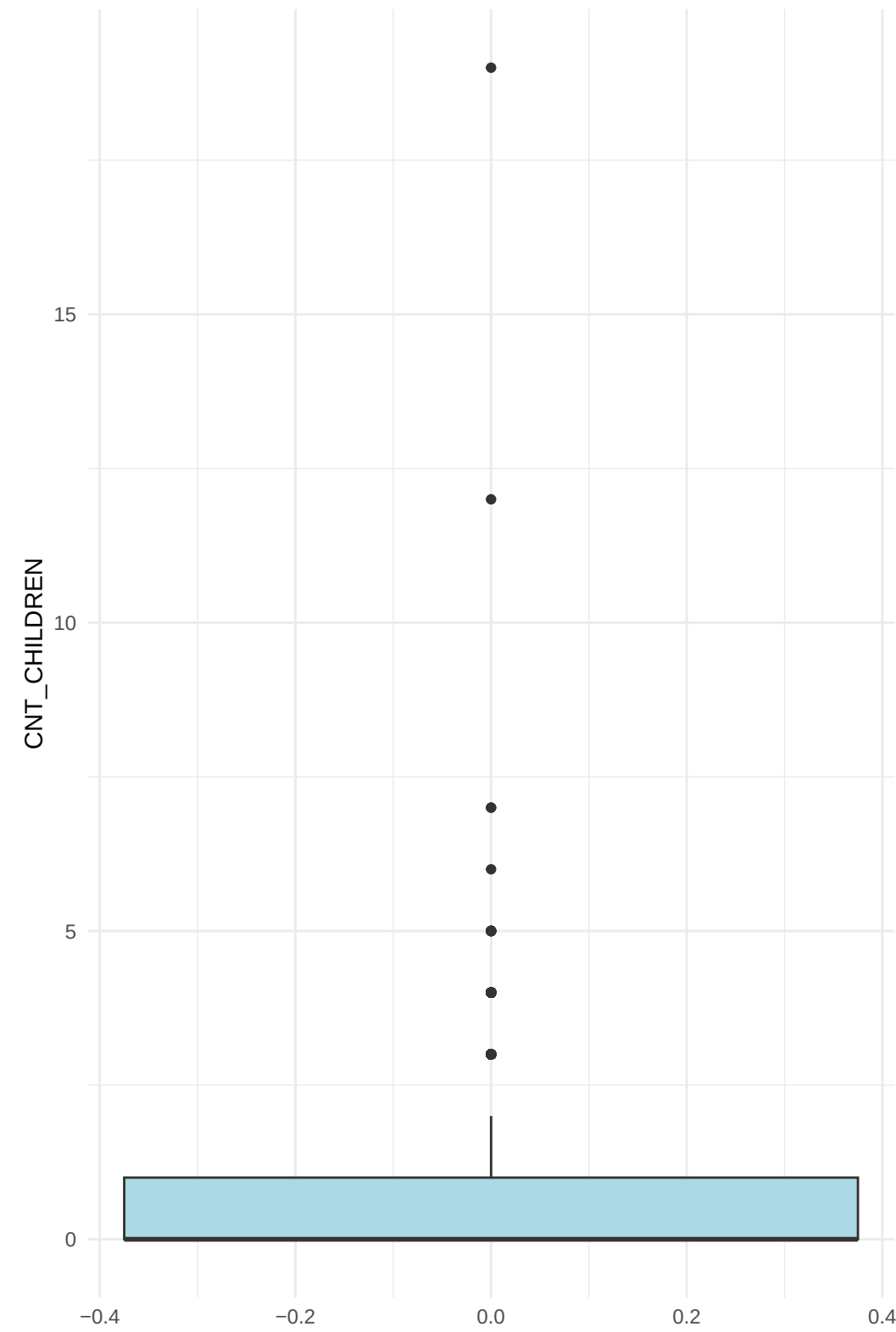
Outliers: ID



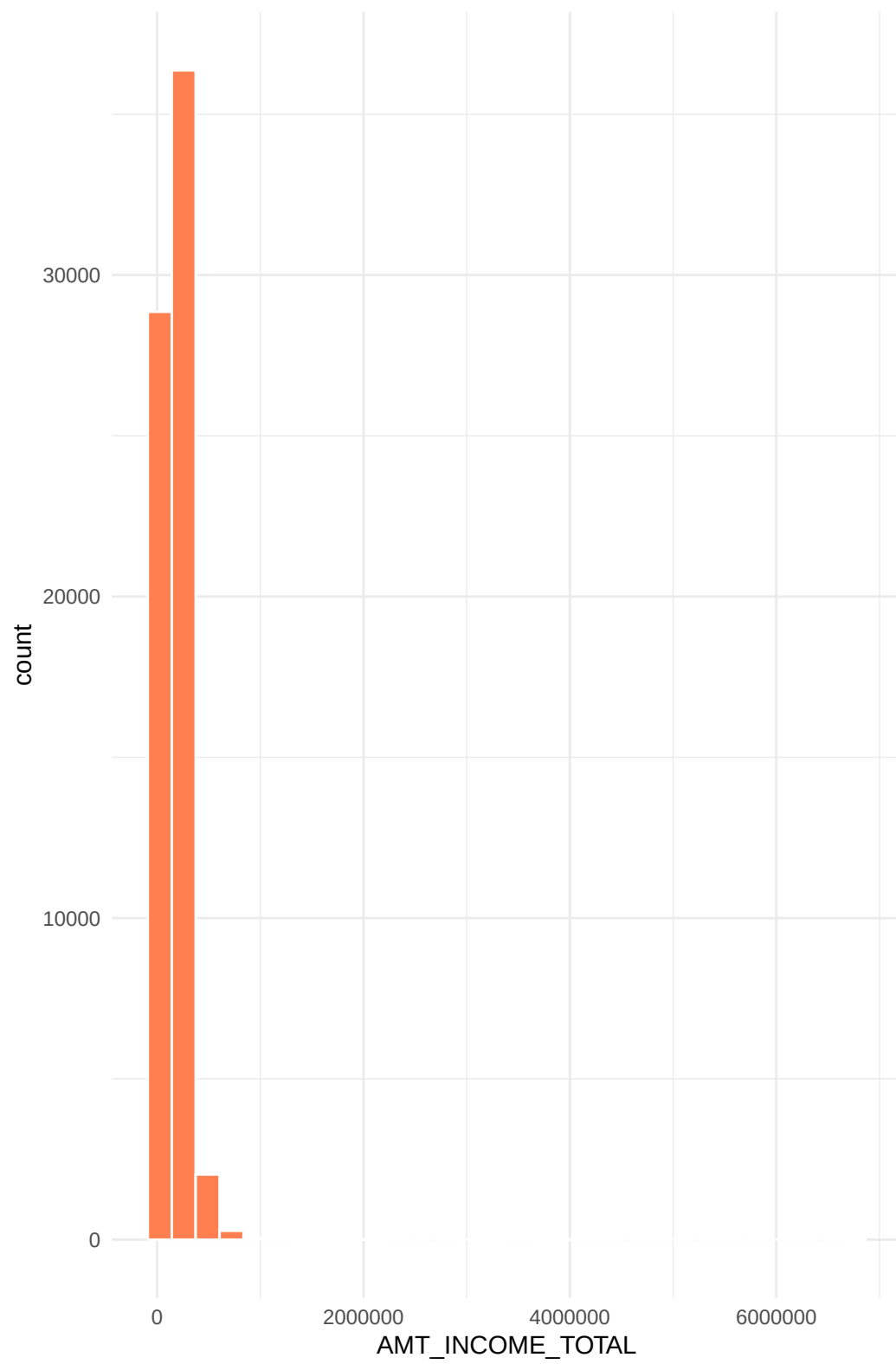
Distribution: CNT_CHILDREN



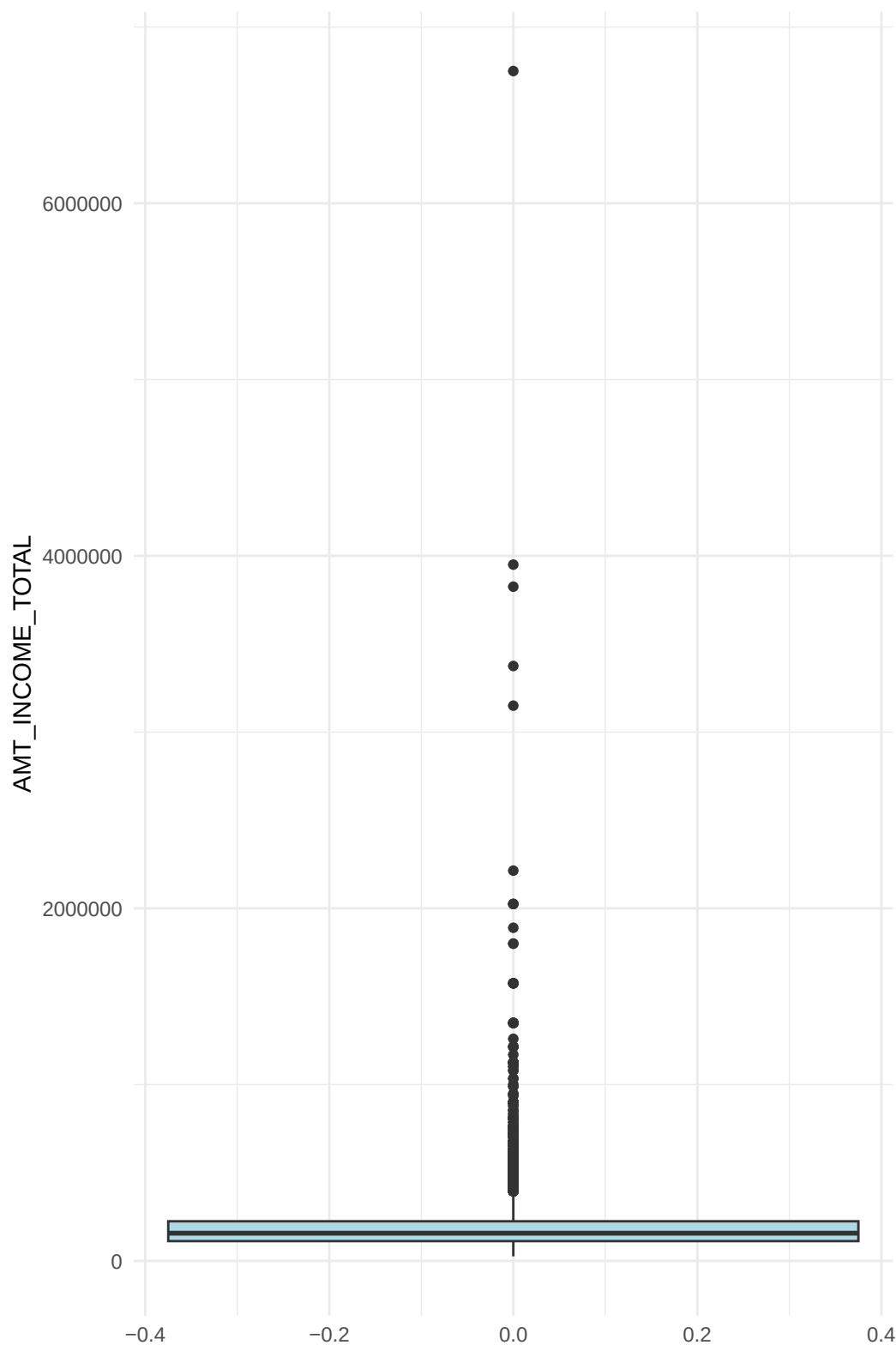
Outliers: CNT_CHILDREN



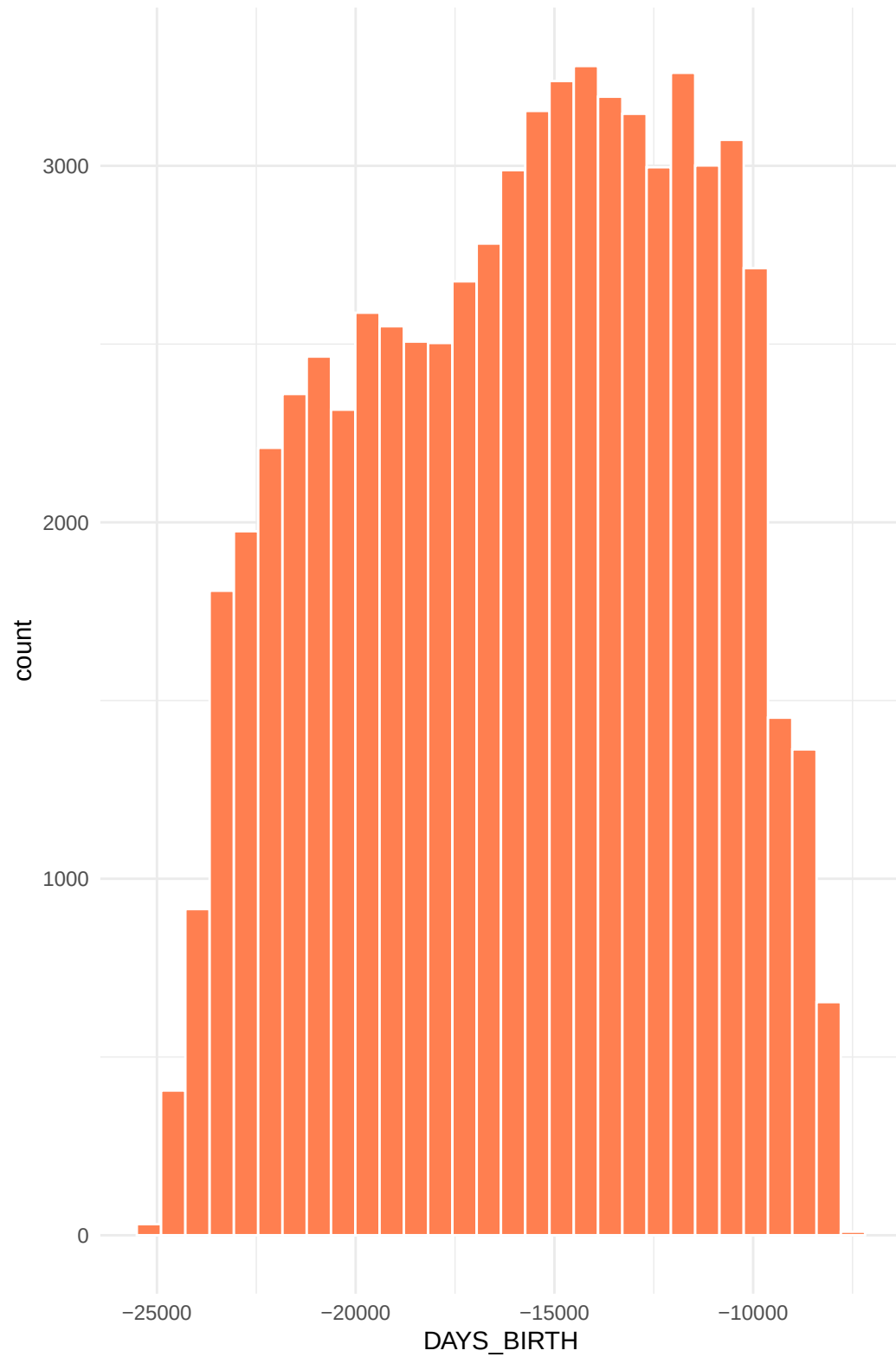
Distribution: AMT_INCOME_TOTAL



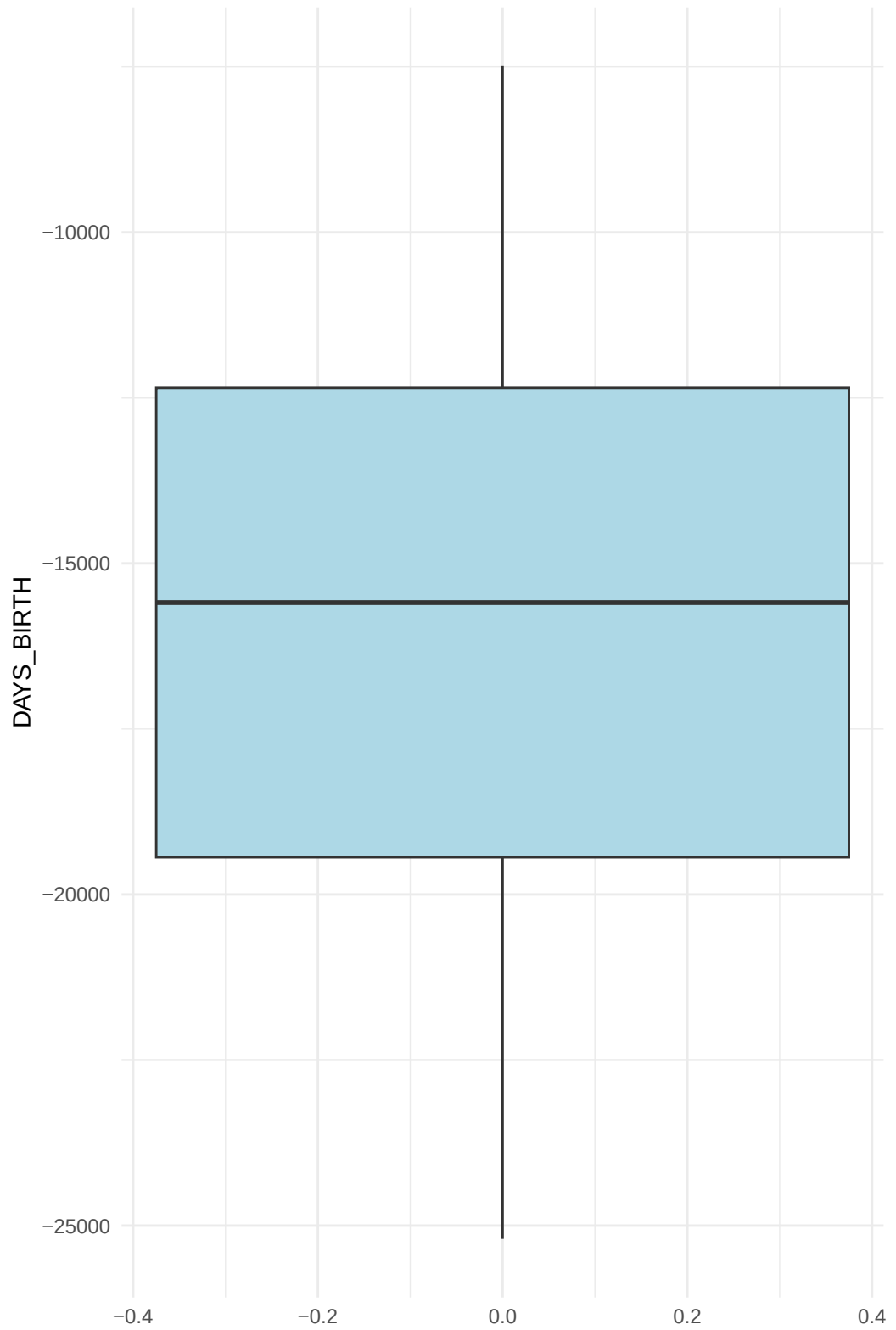
Outliers: AMT_INCOME_TOTAL



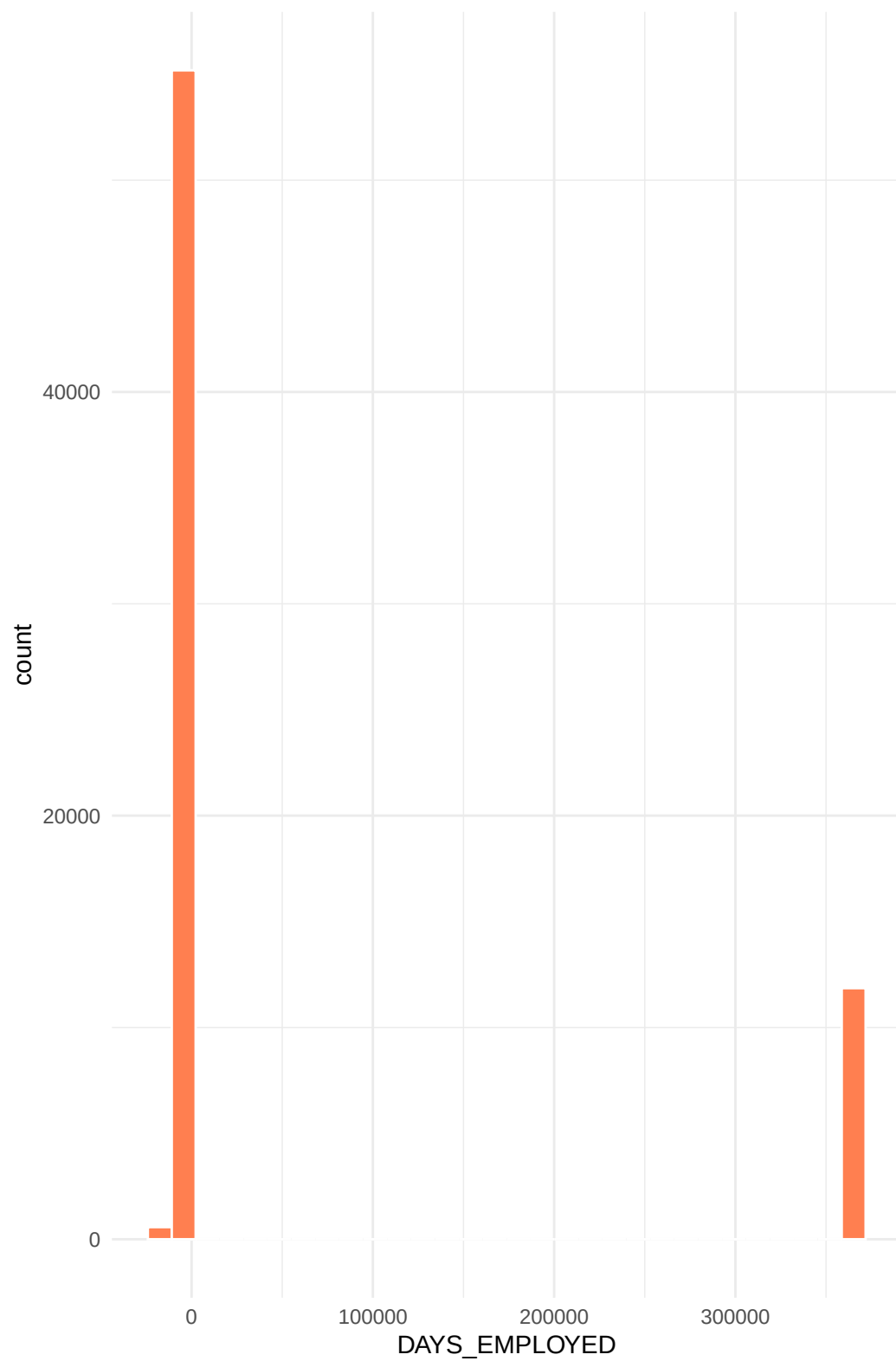
Distribution: DAYS_BIRTH



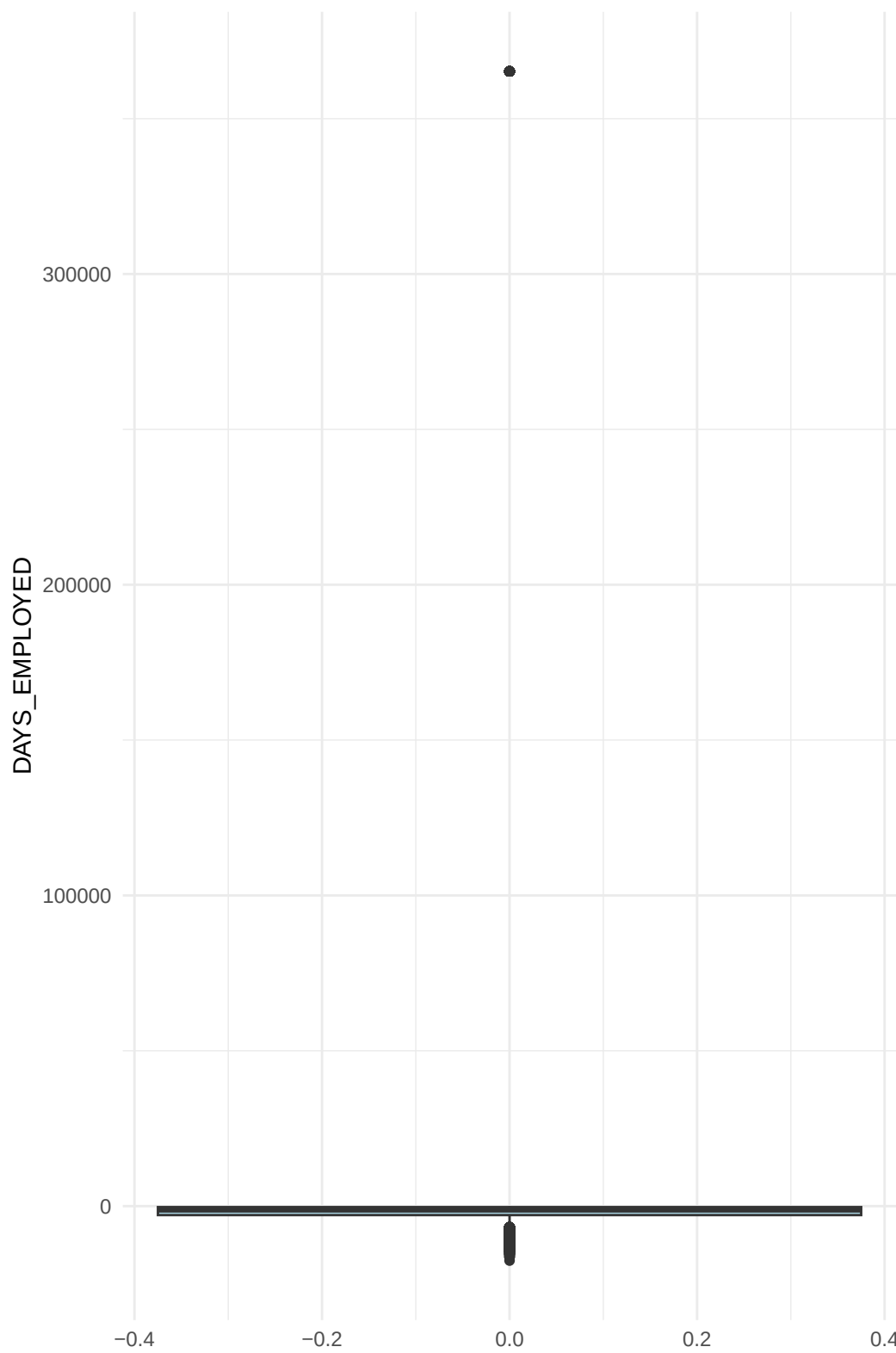
Outliers: DAYS_BIRTH



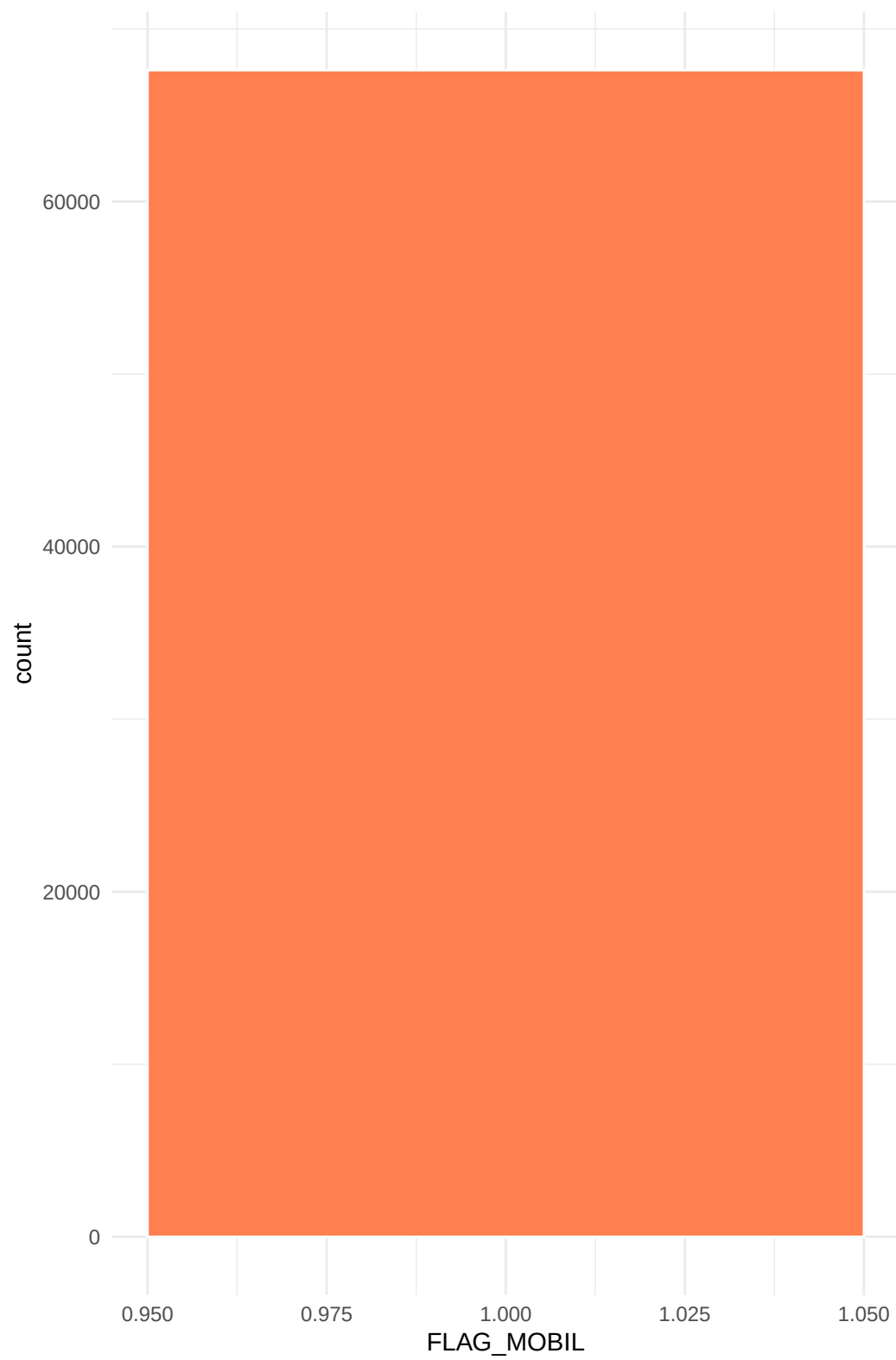
Distribution: DAYS_EMPLOYED



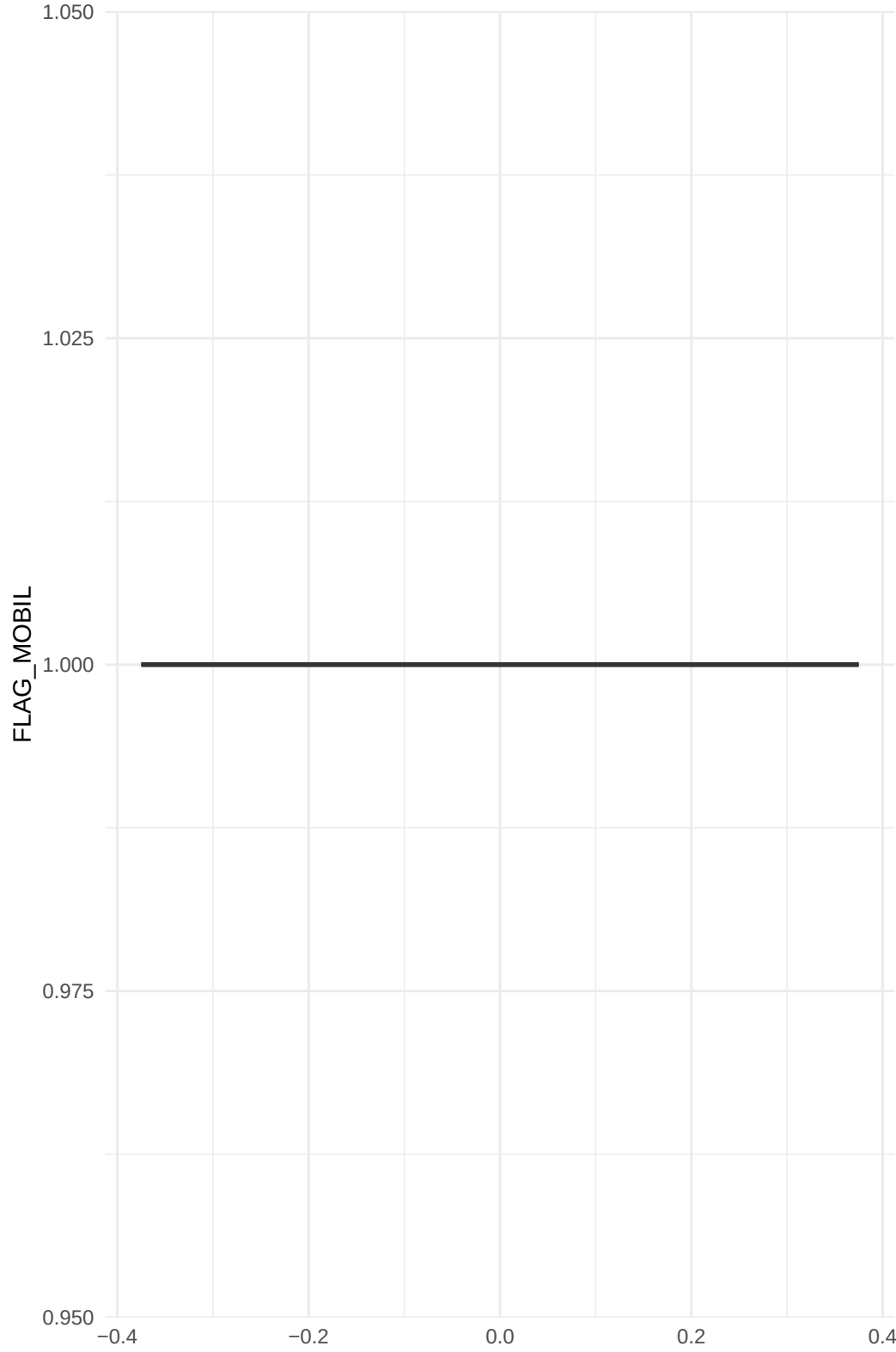
Outliers: DAYS_EMPLOYED



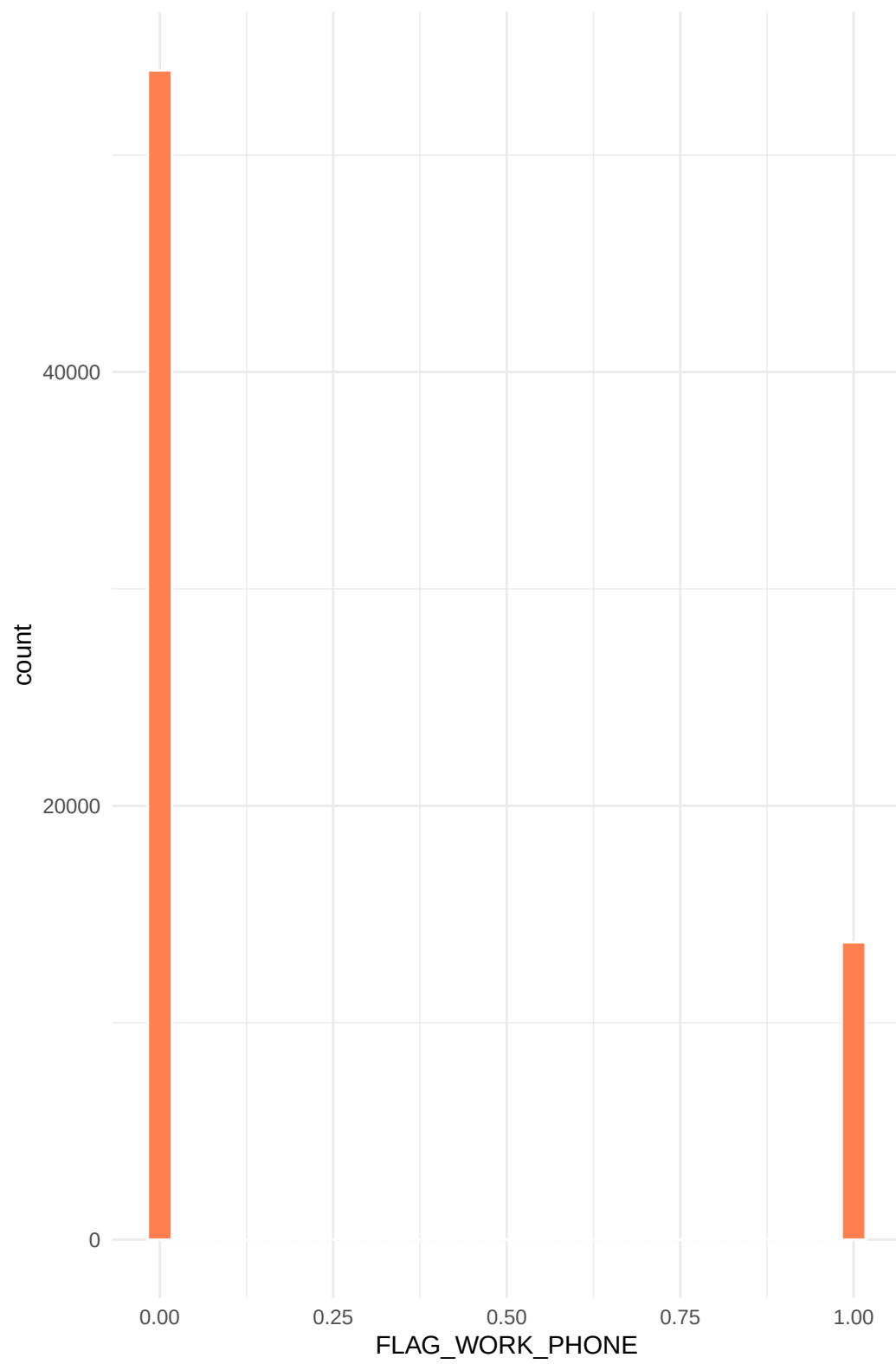
Distribution: FLAG_MOBIL



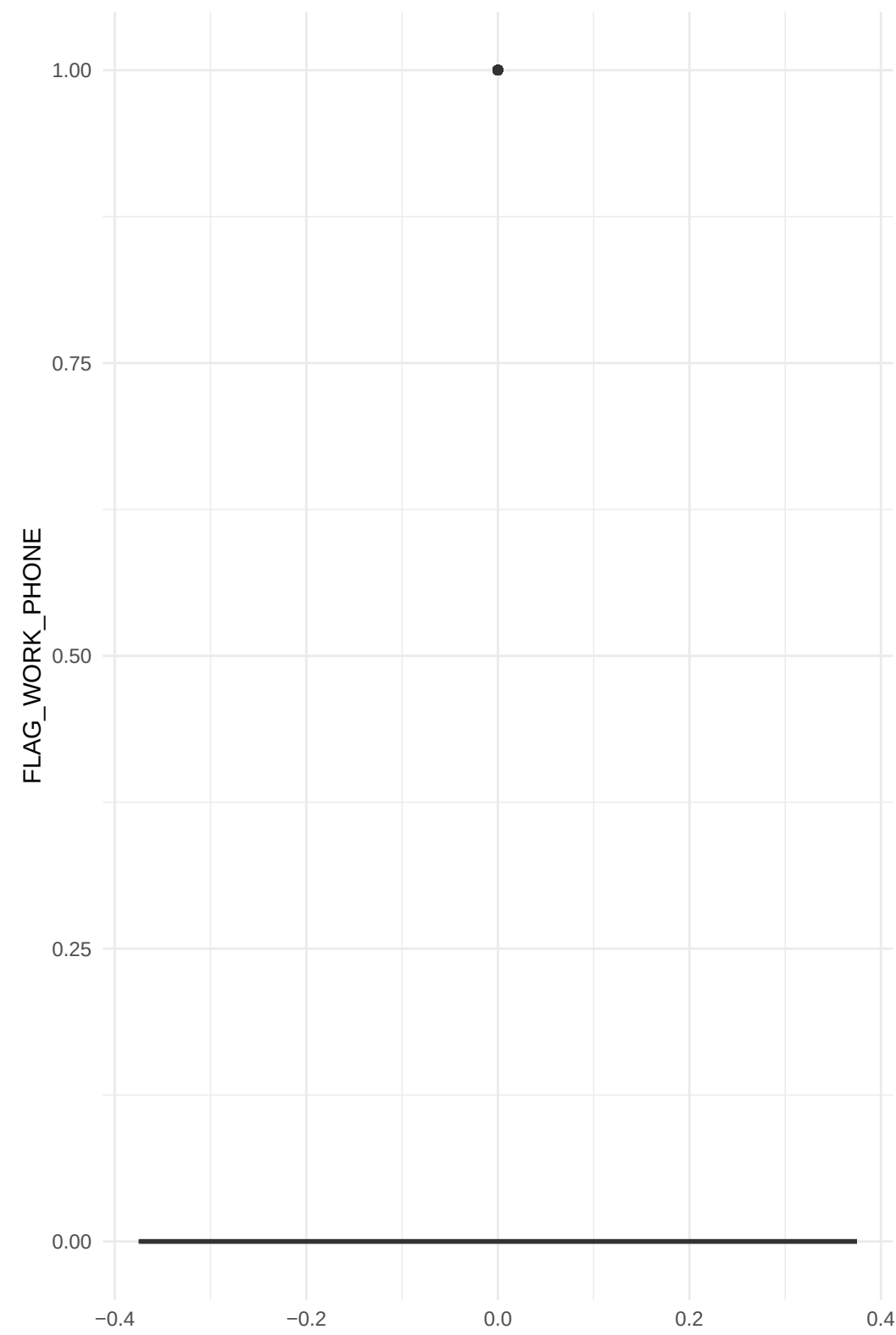
Outliers: FLAG_MOBIL



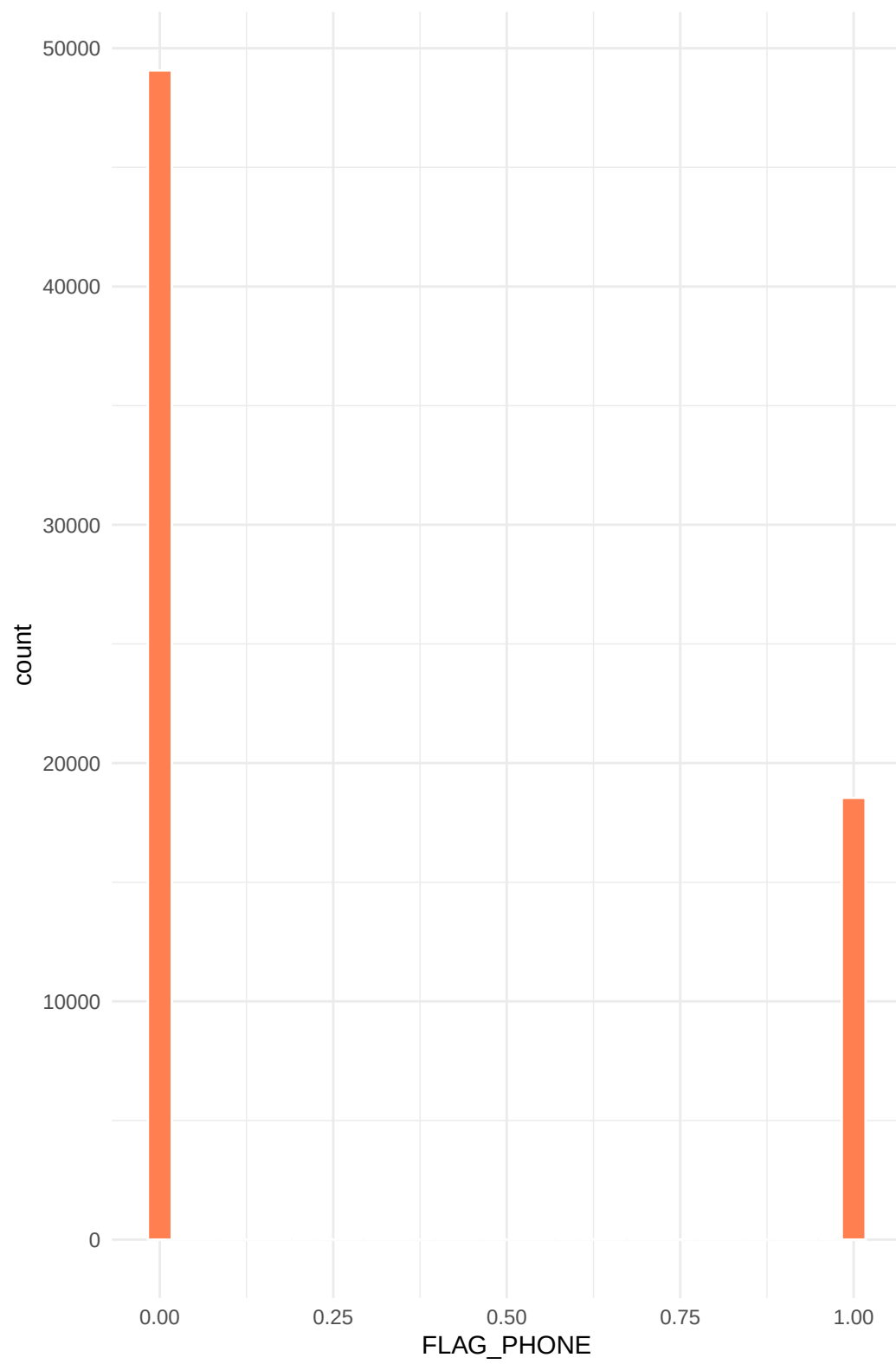
Distribution: FLAG_WORK_PHONE



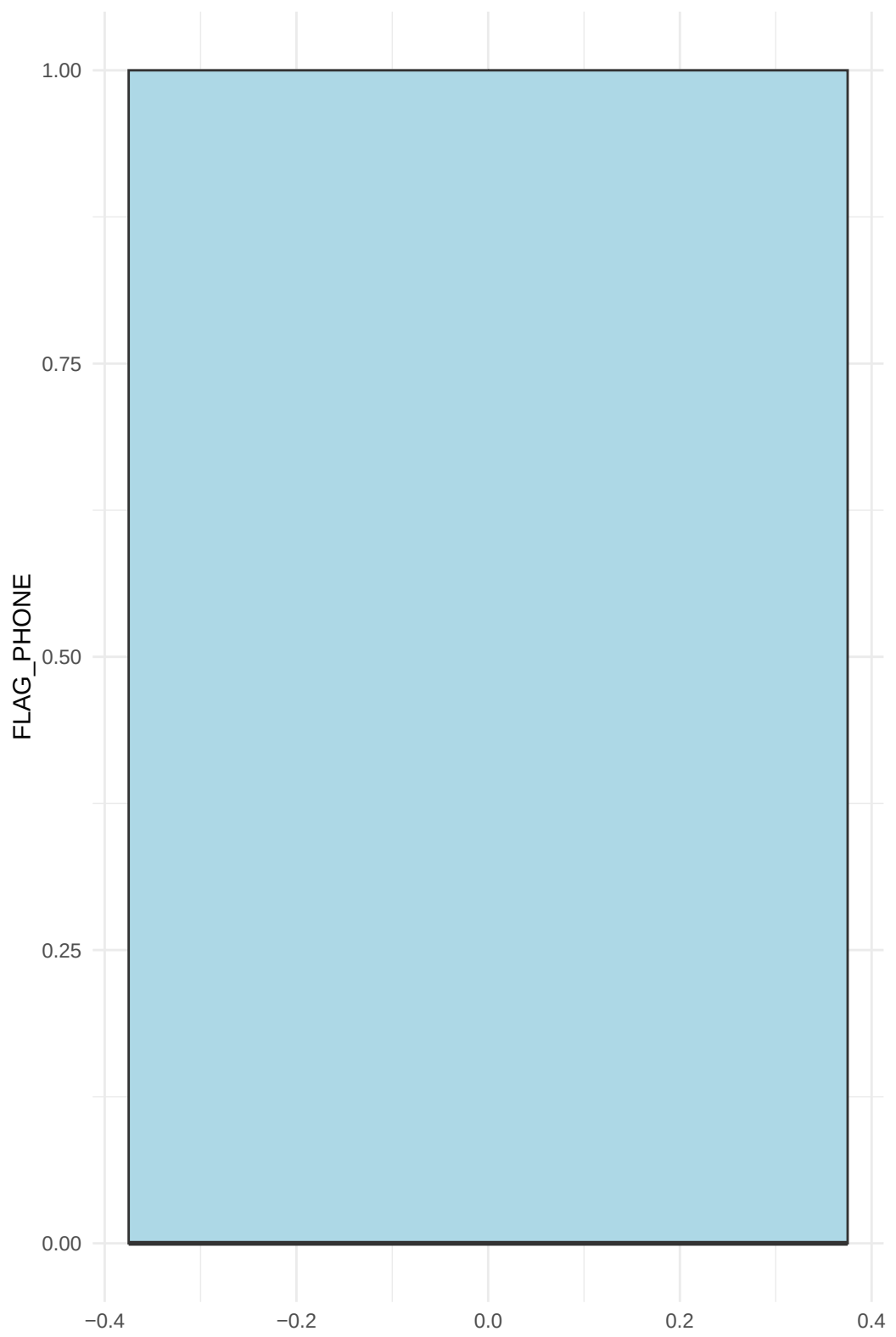
Outliers: FLAG_WORK_PHONE



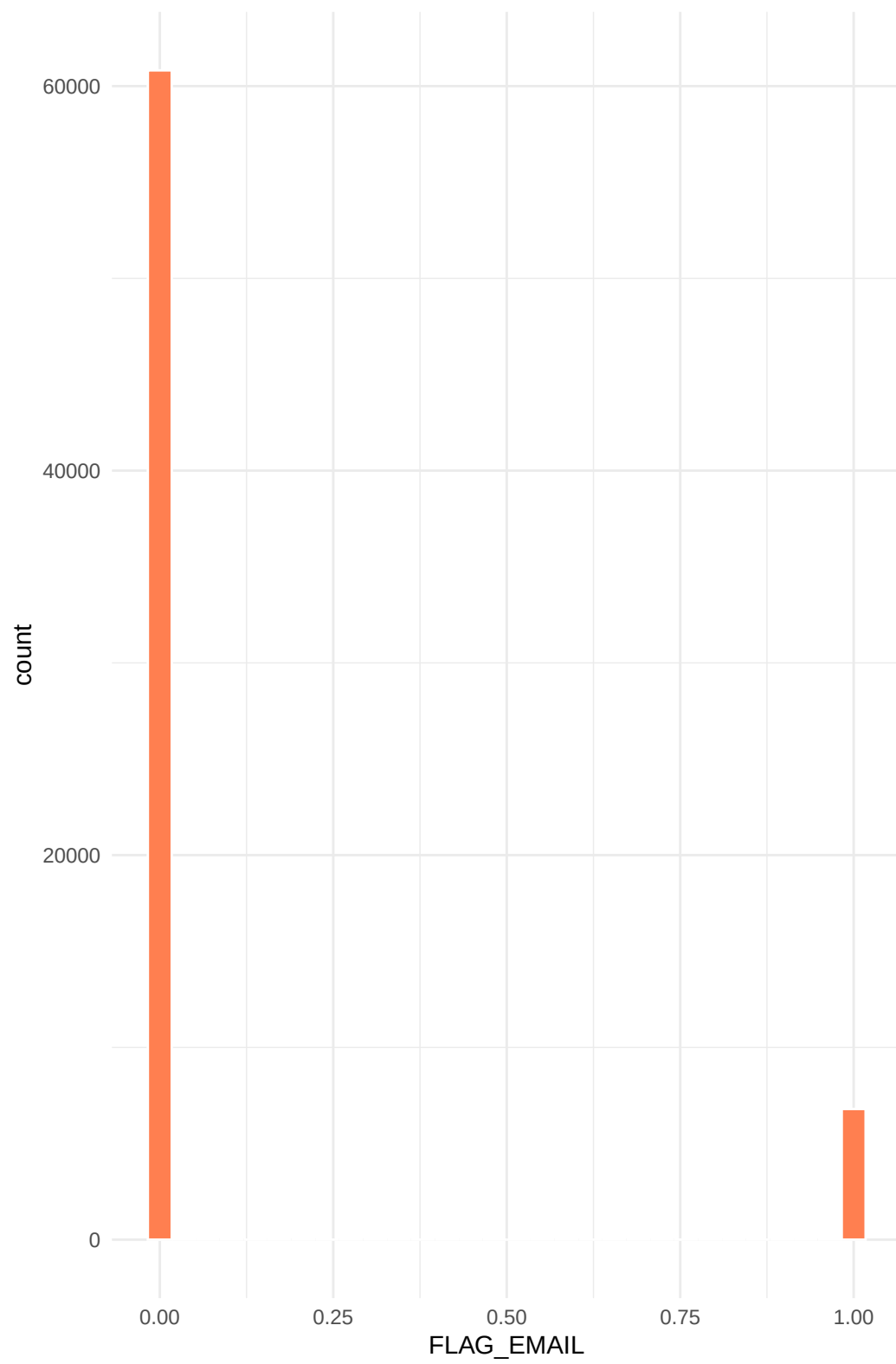
Distribution: FLAG_PHONE



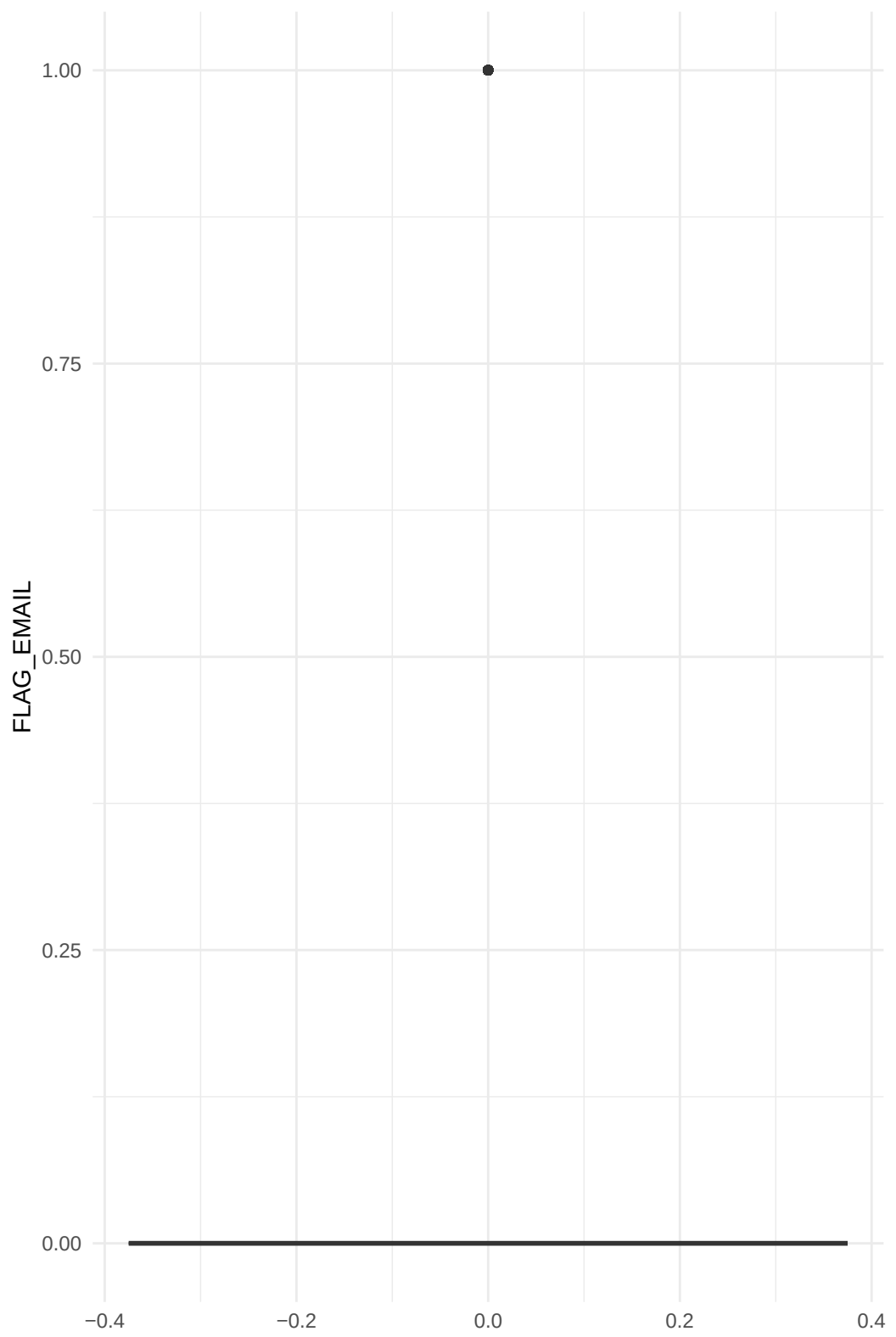
Outliers: FLAG_PHONE



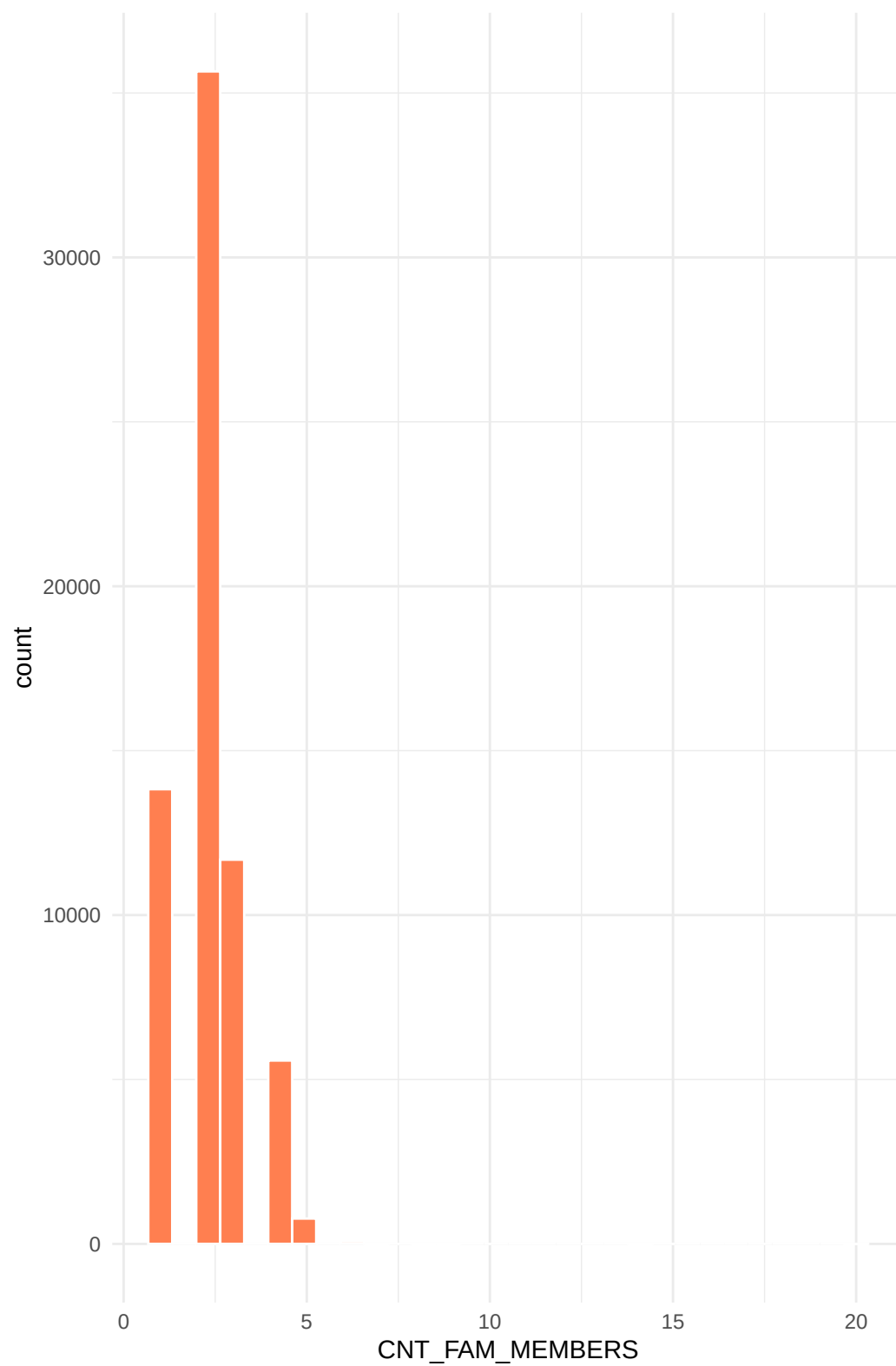
Distribution: FLAG_EMAIL



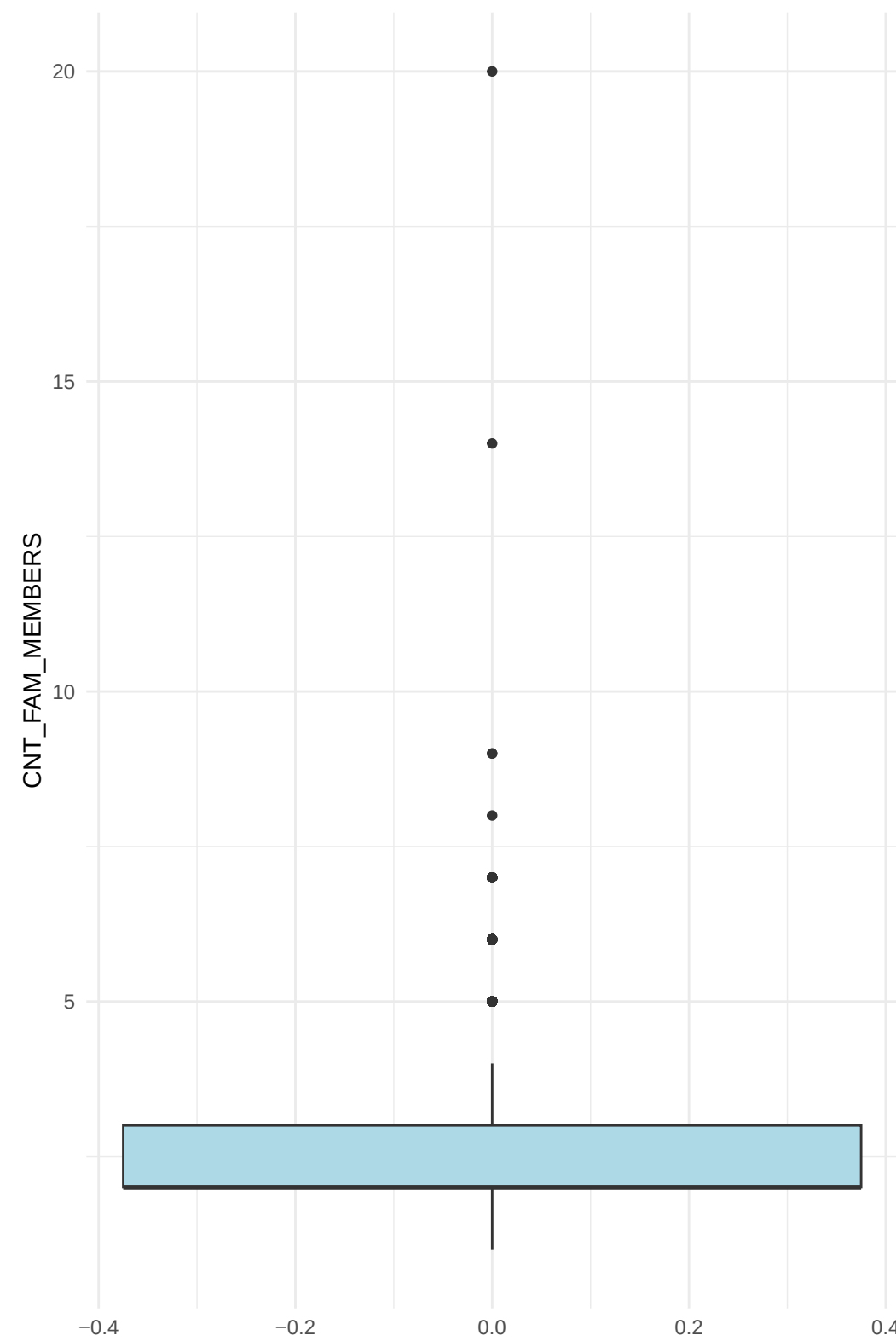
Outliers: FLAG_EMAIL



Distribution: CNT_FAM_MEMBERS

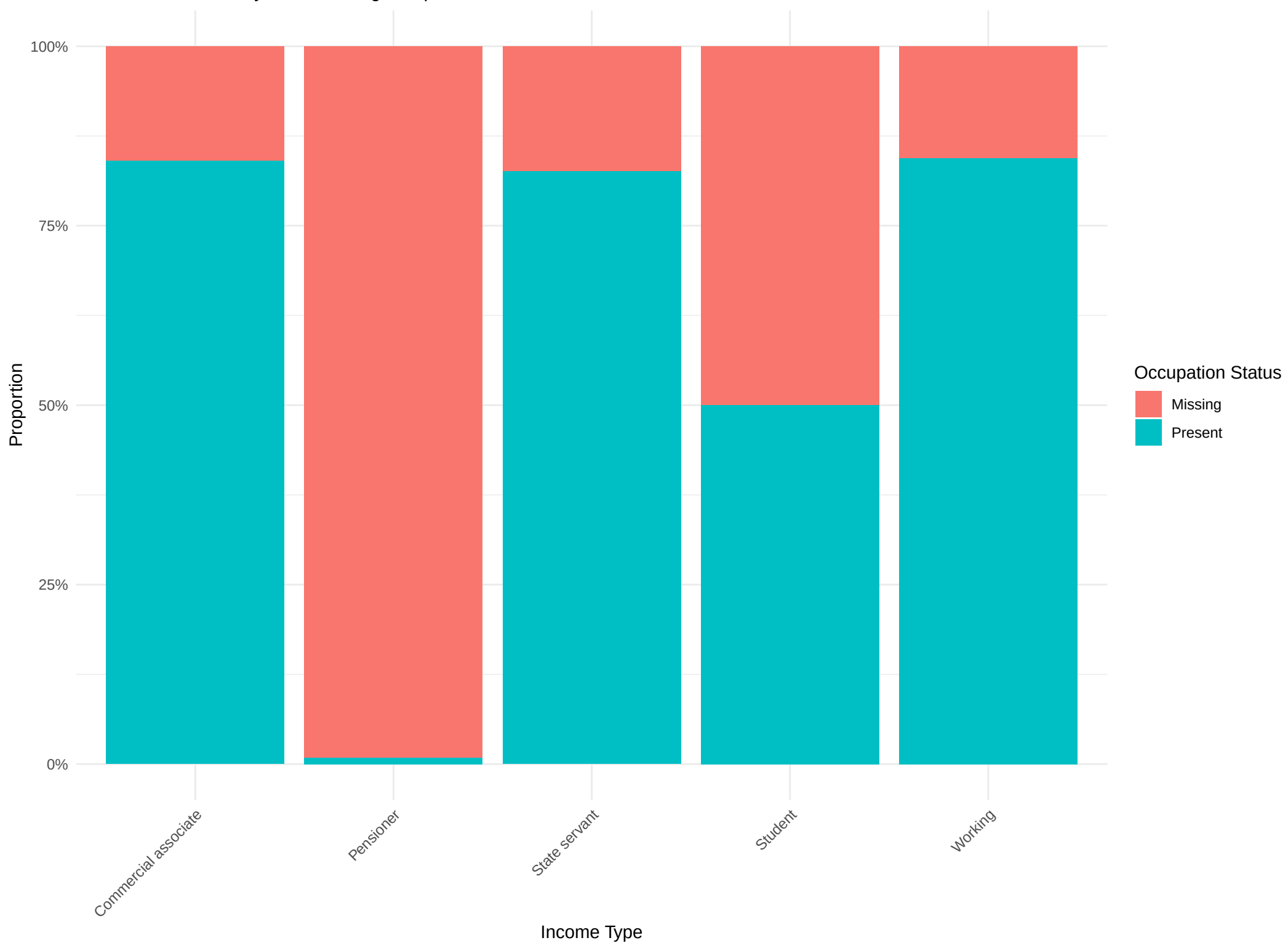


Outliers: CNT_FAM_MEMBERS

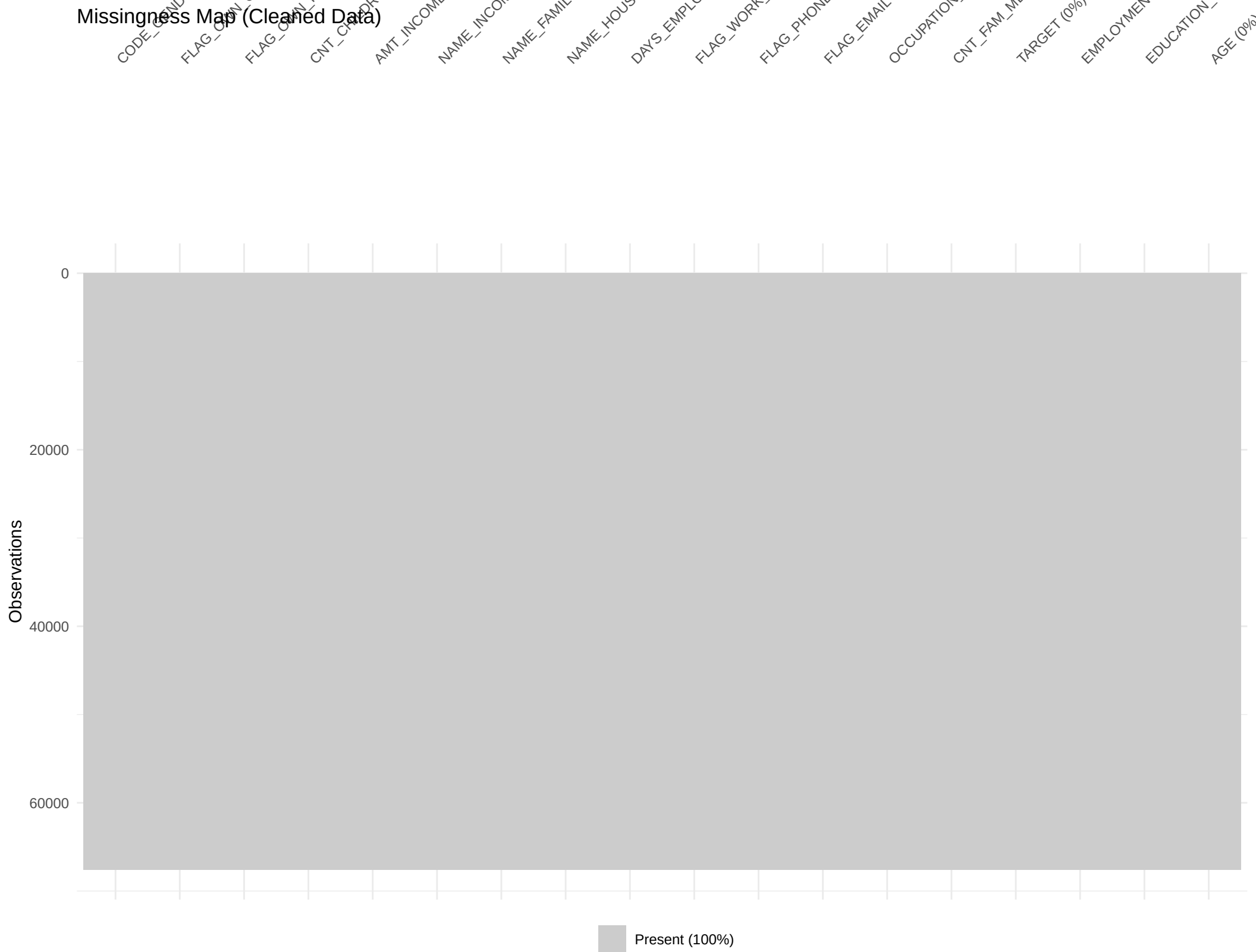


Evidence of MNAR: Occupation Missingness by Income Type

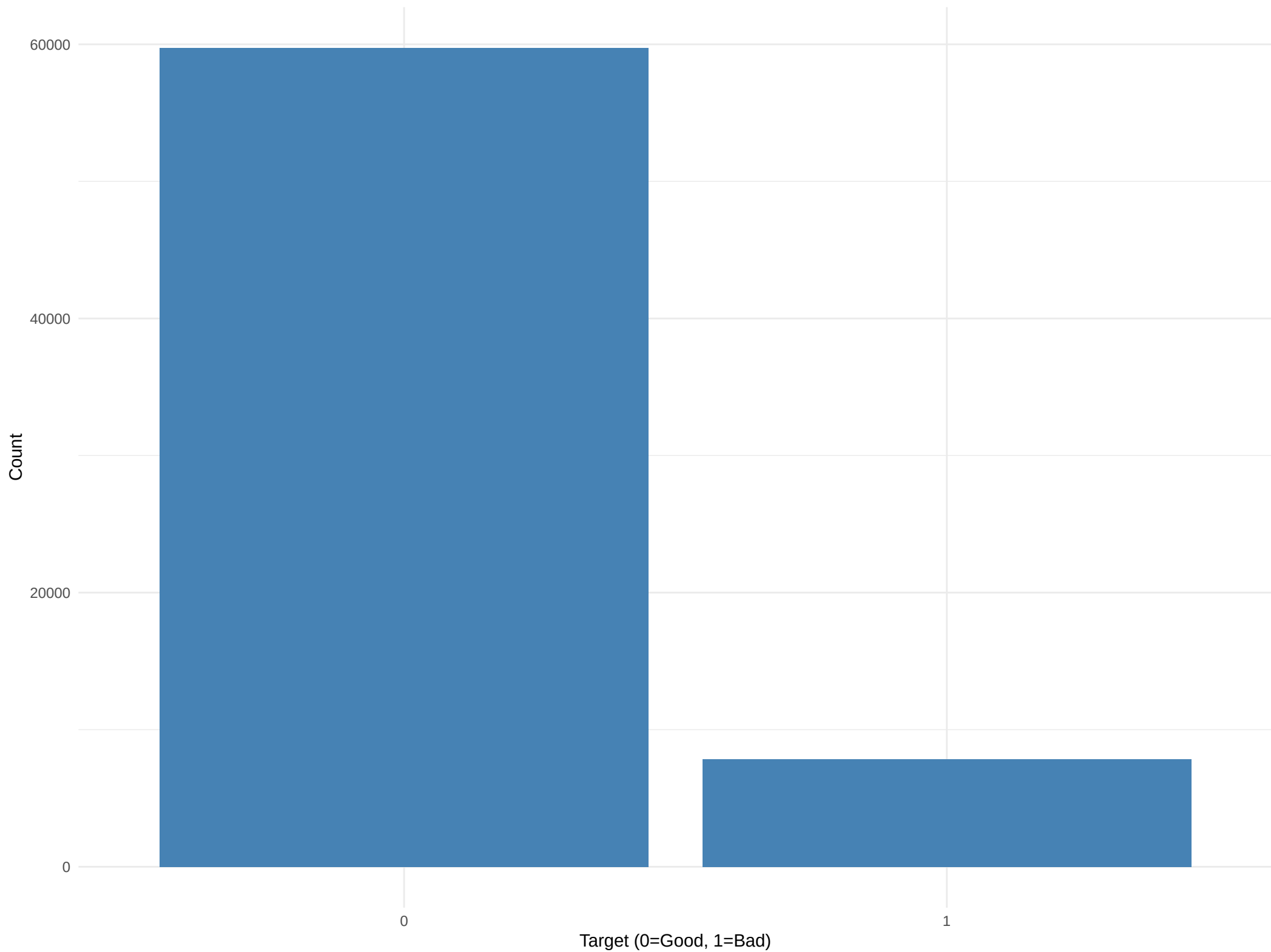
Pensioners have nearly 100% missing occupation data



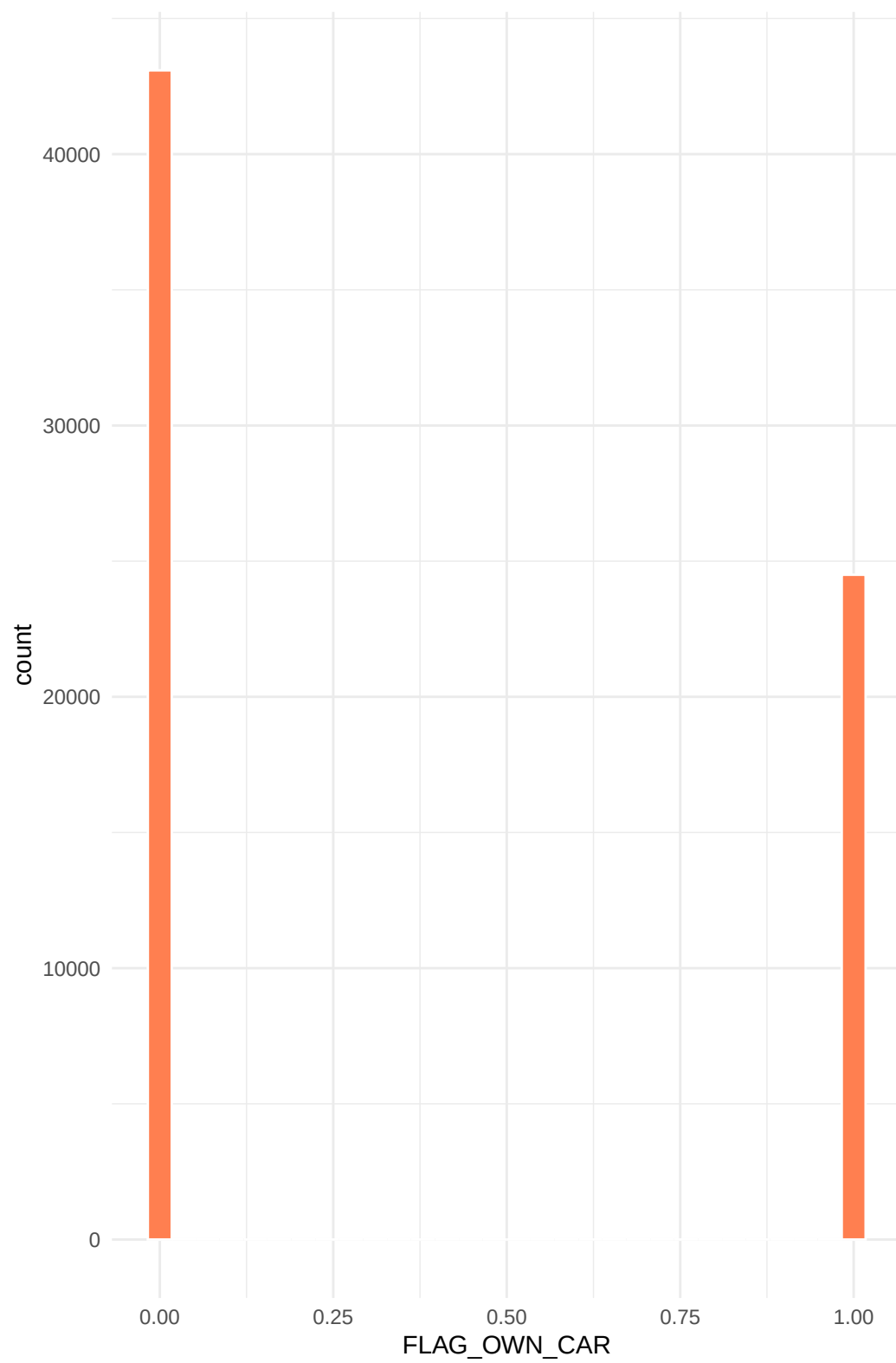
Missingness Map (Cleaned Data)



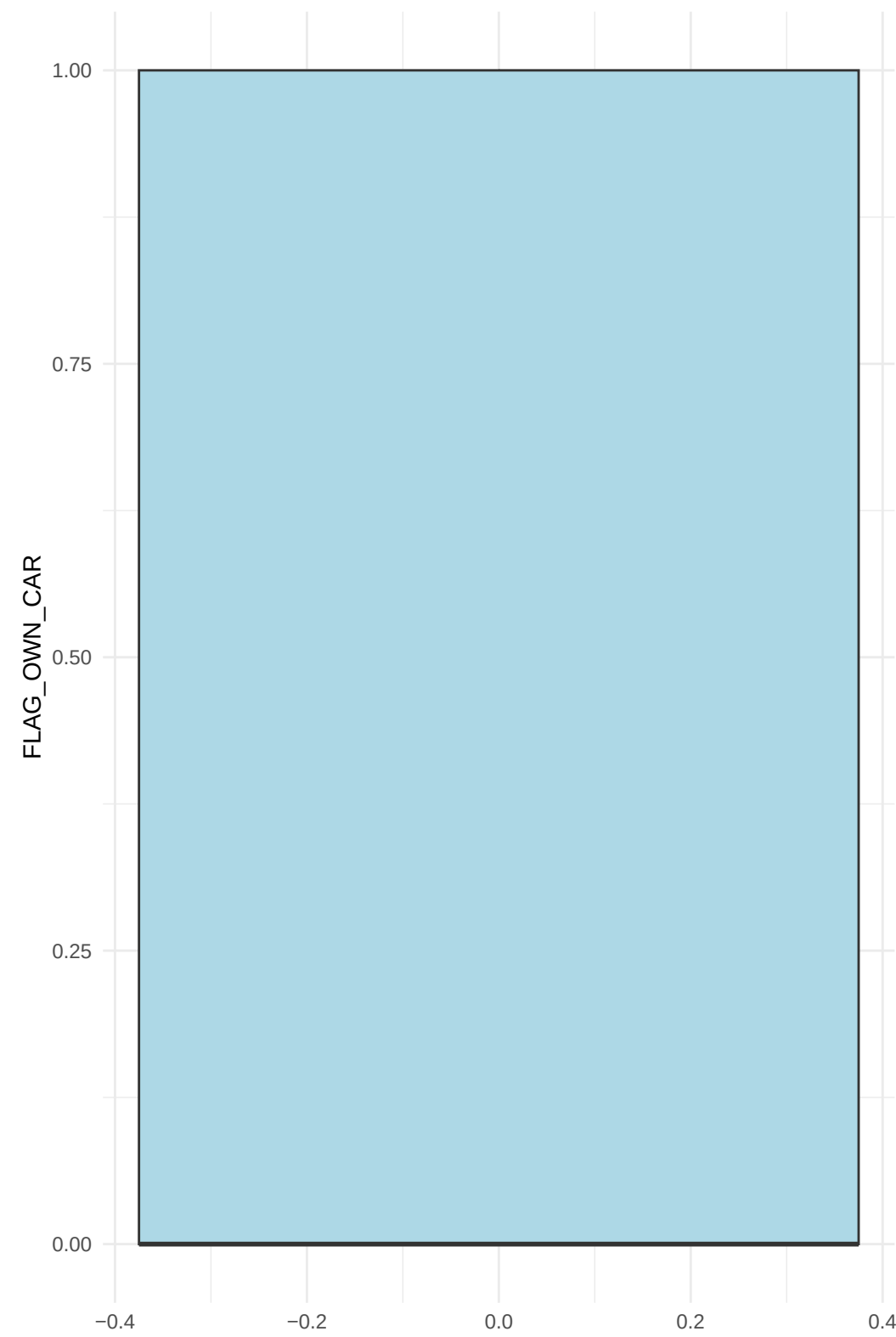
Final Target Distribution



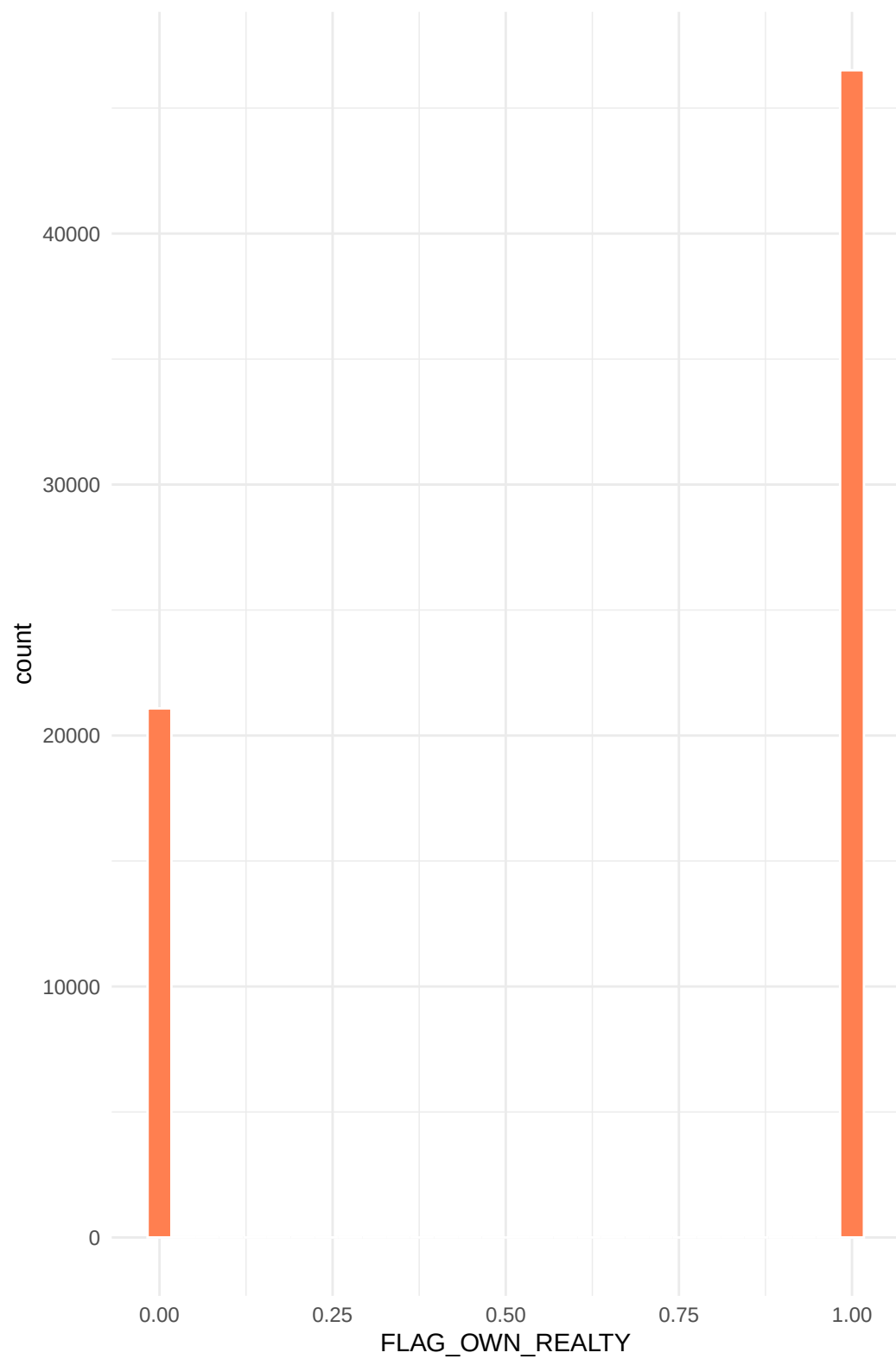
Distribution: FLAG_OWN_CAR



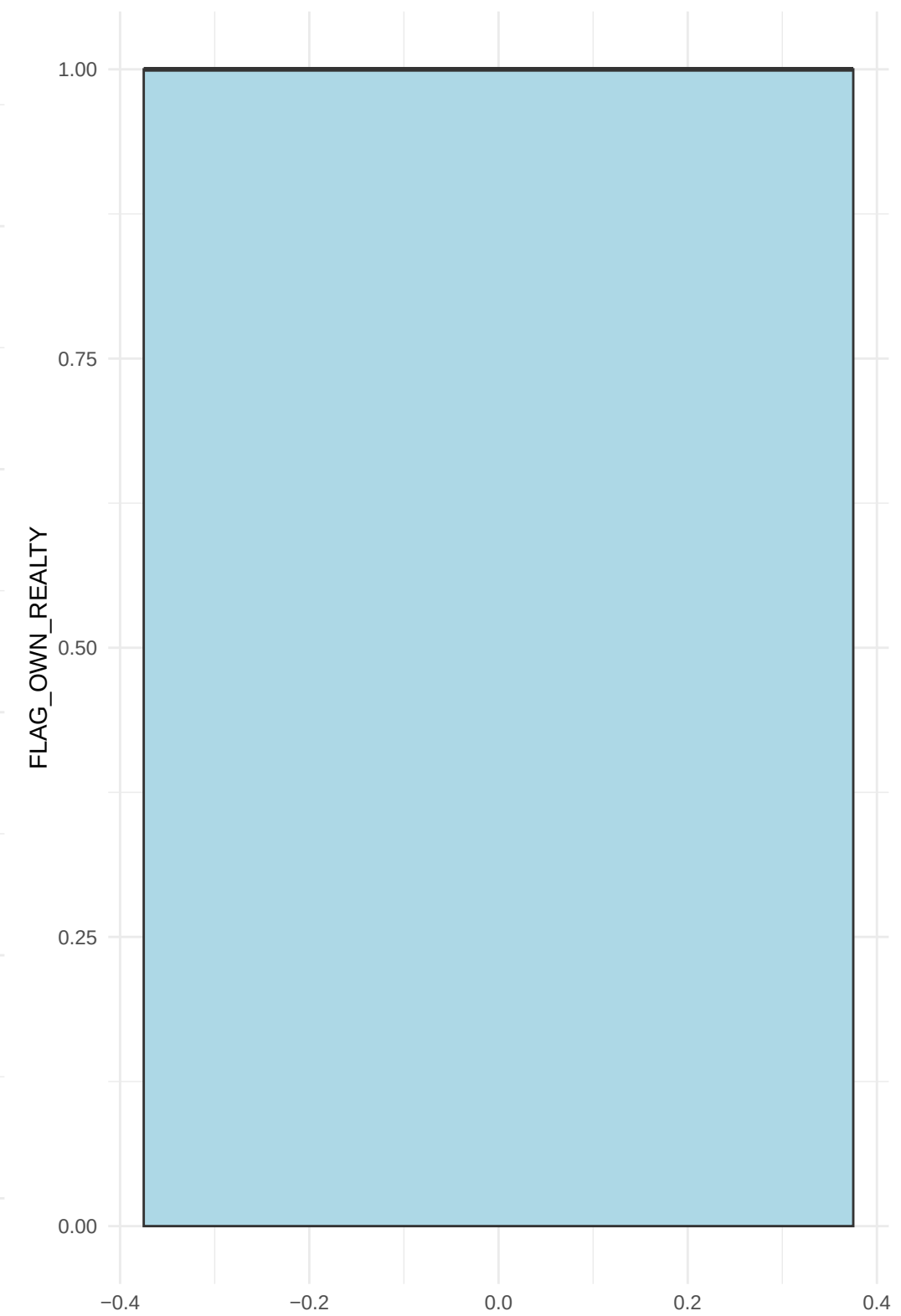
Outliers: FLAG_OWN_CAR



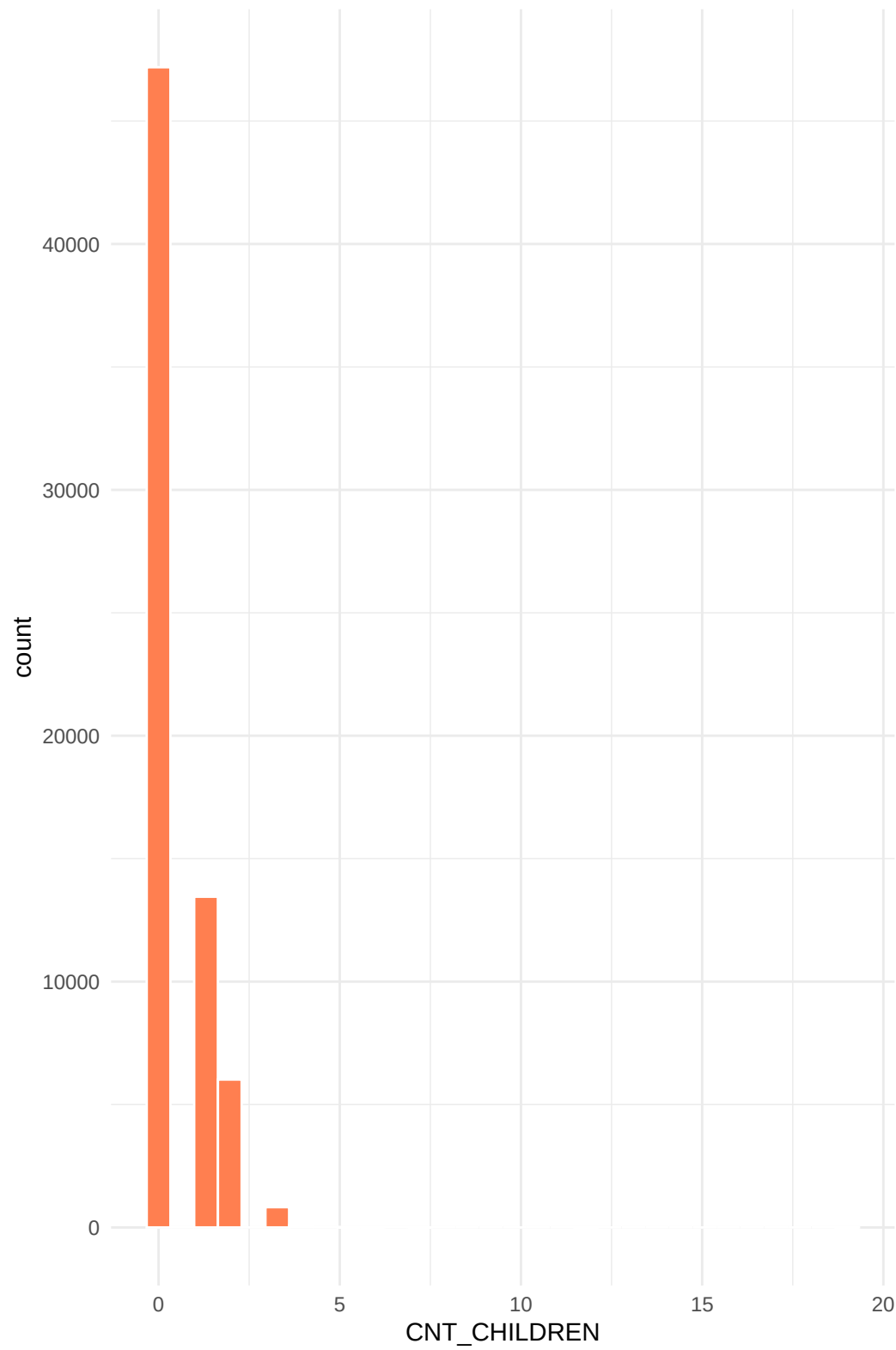
Distribution: FLAG_OWN_REALTY



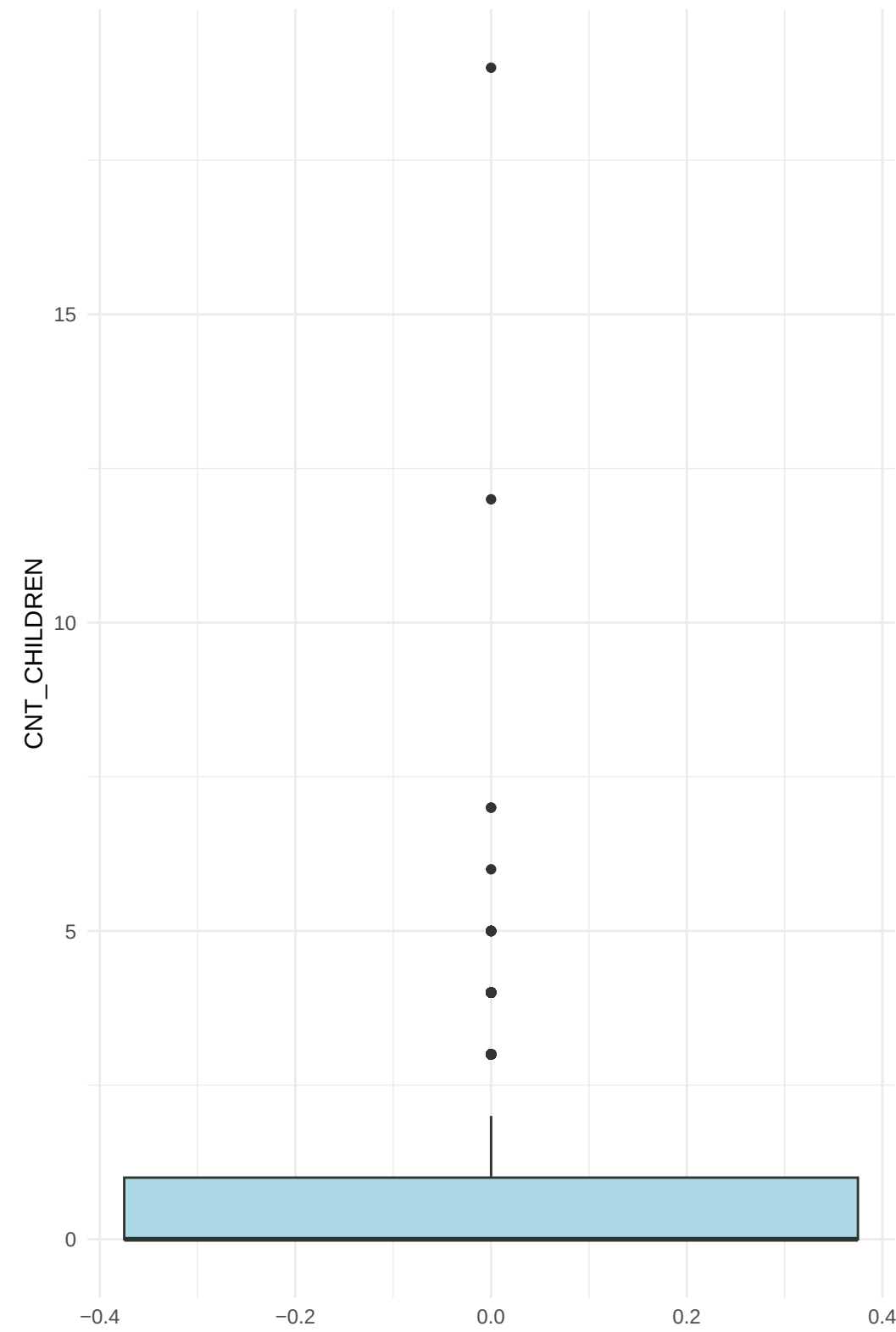
Outliers: FLAG_OWN_REALTY



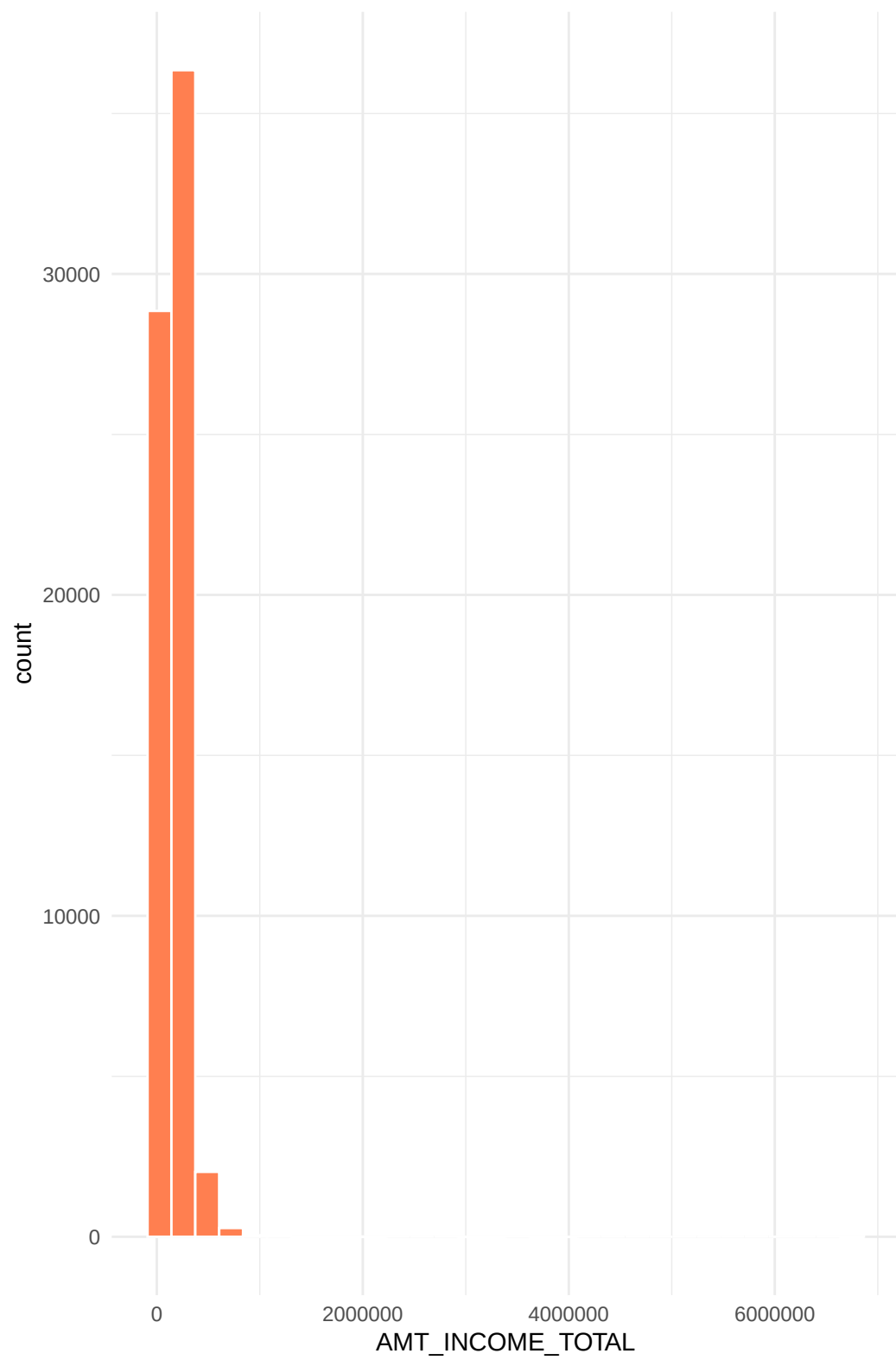
Distribution: CNT_CHILDREN



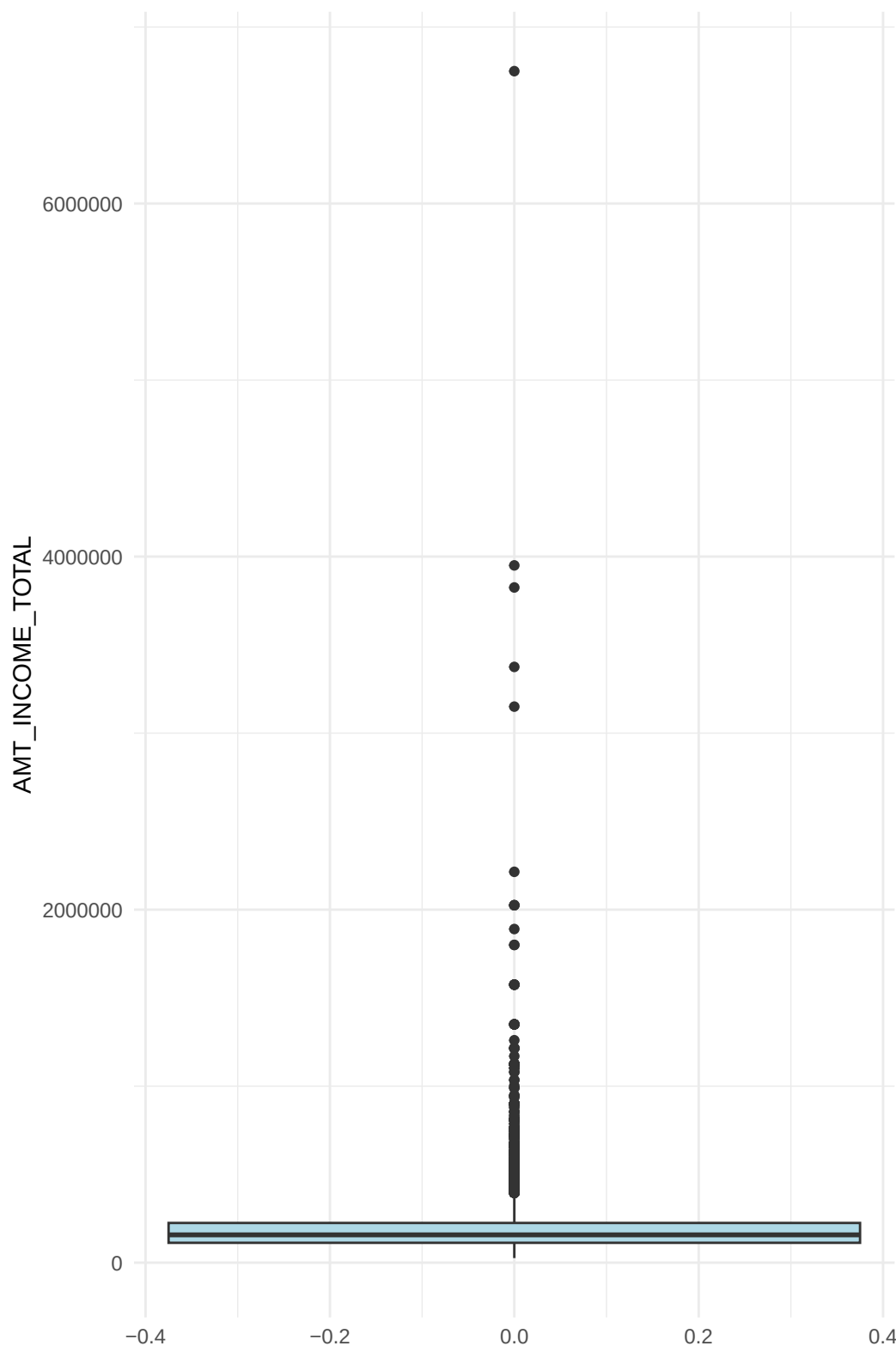
Outliers: CNT_CHILDREN



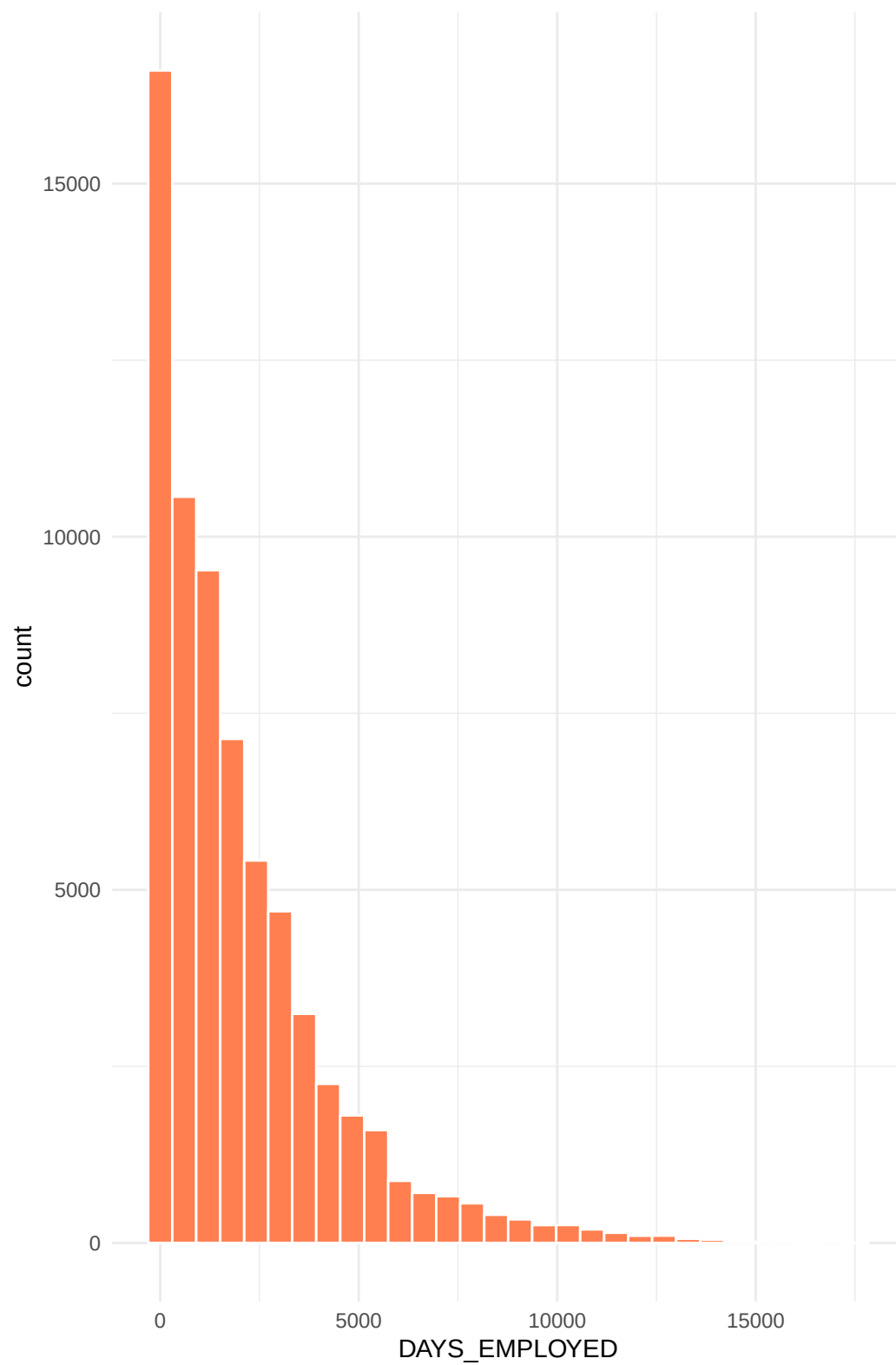
Distribution: AMT_INCOME_TOTAL



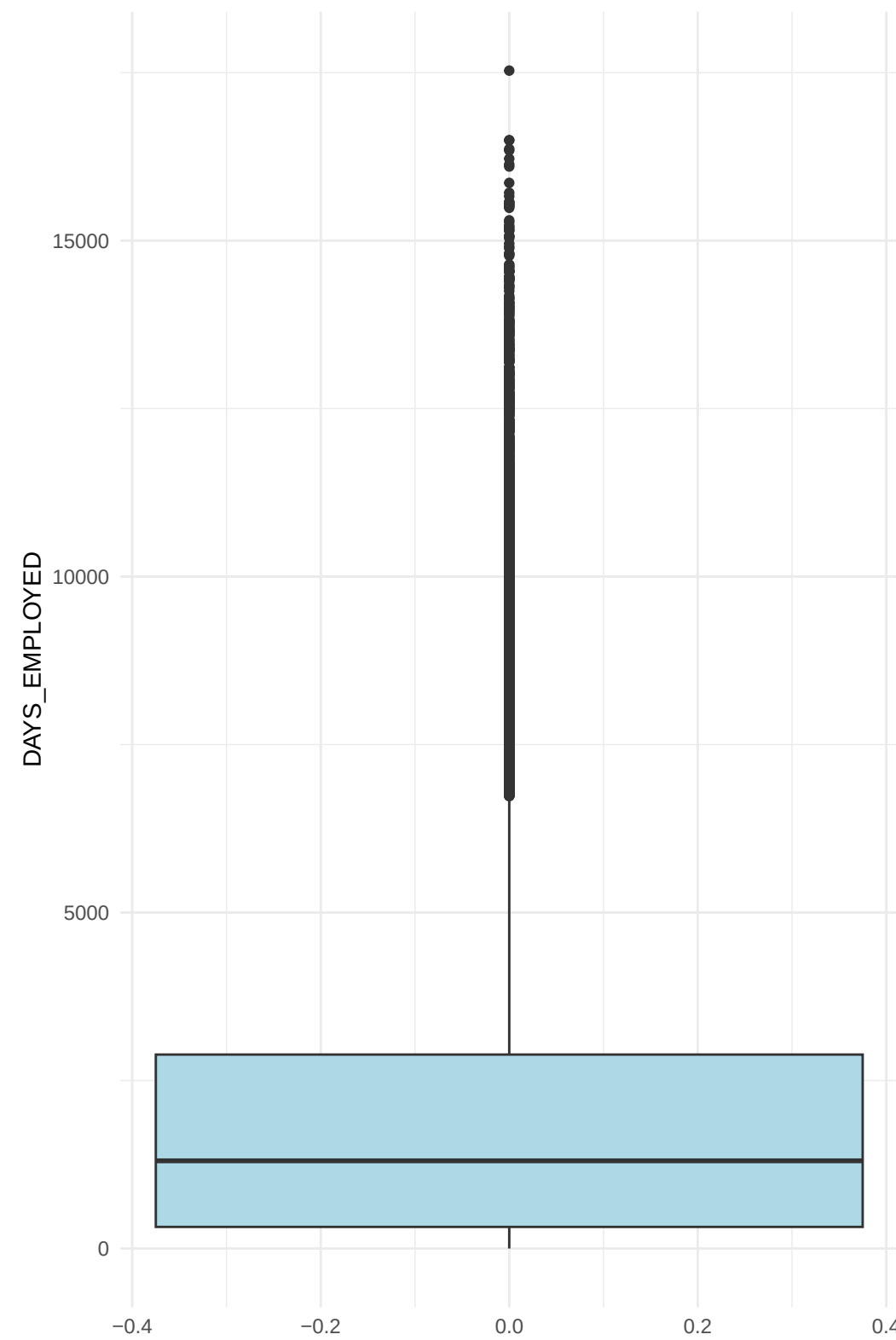
Outliers: AMT_INCOME_TOTAL



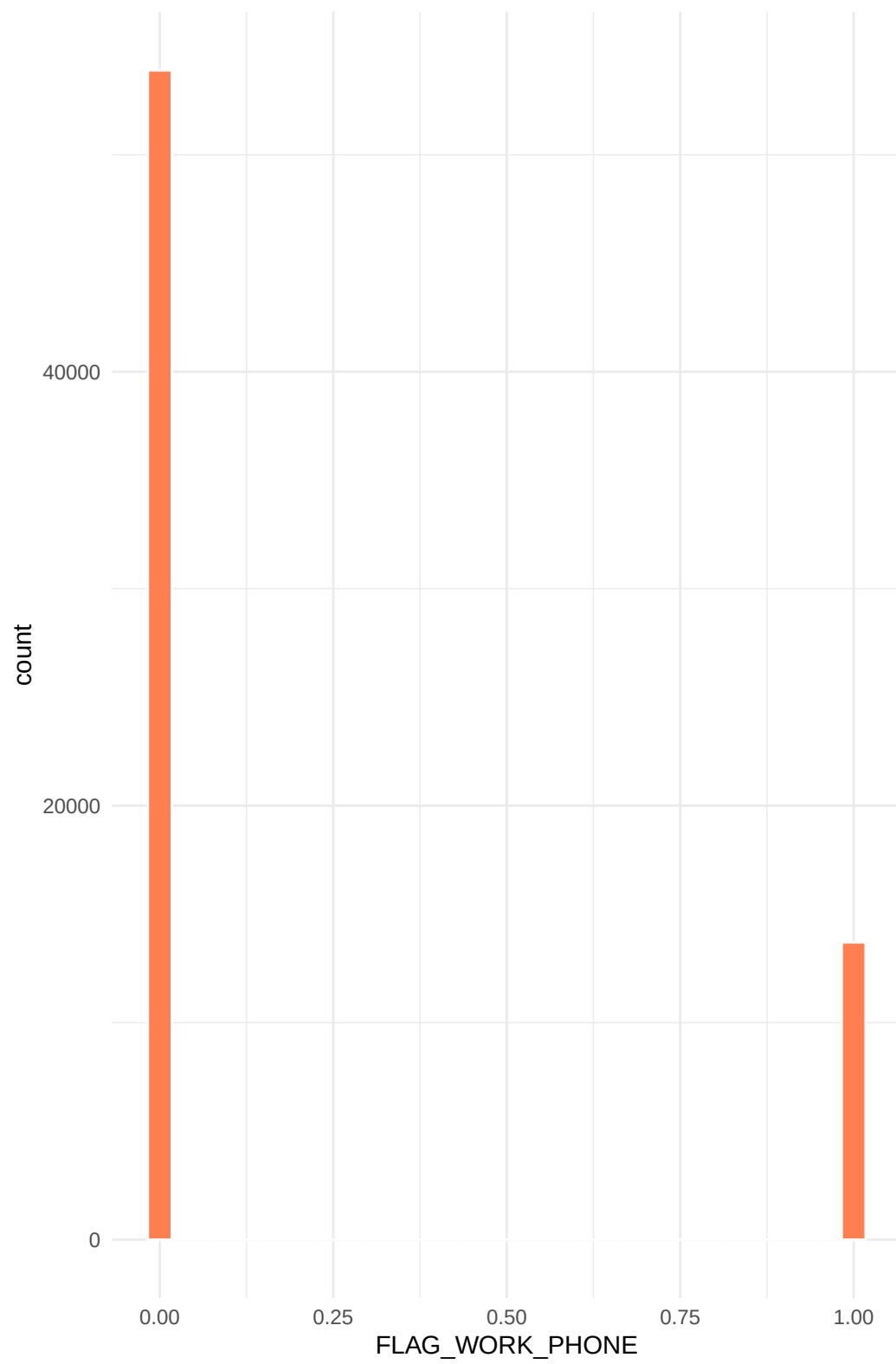
Distribution: DAYS_EMPLOYED



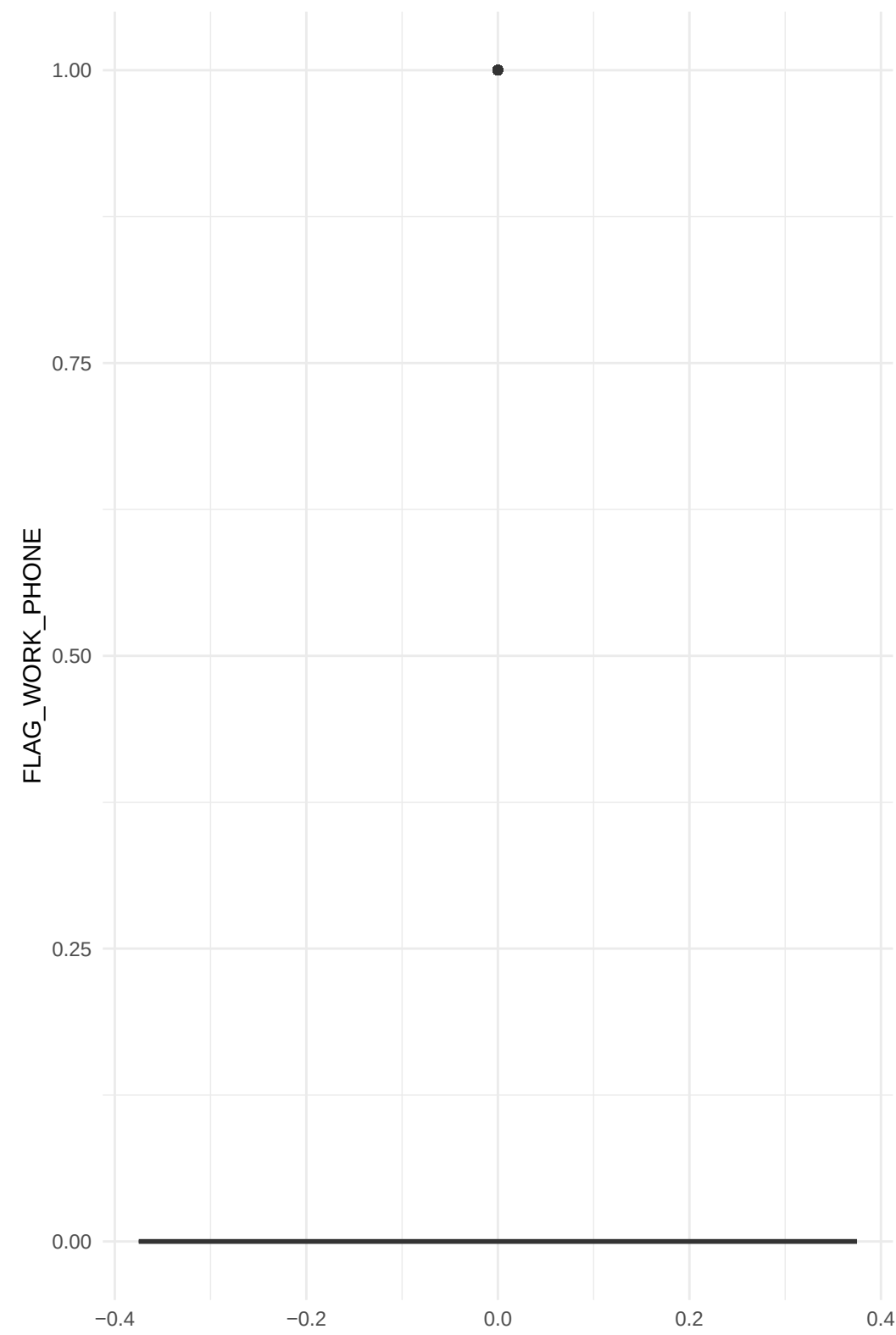
Outliers: DAYS_EMPLOYED



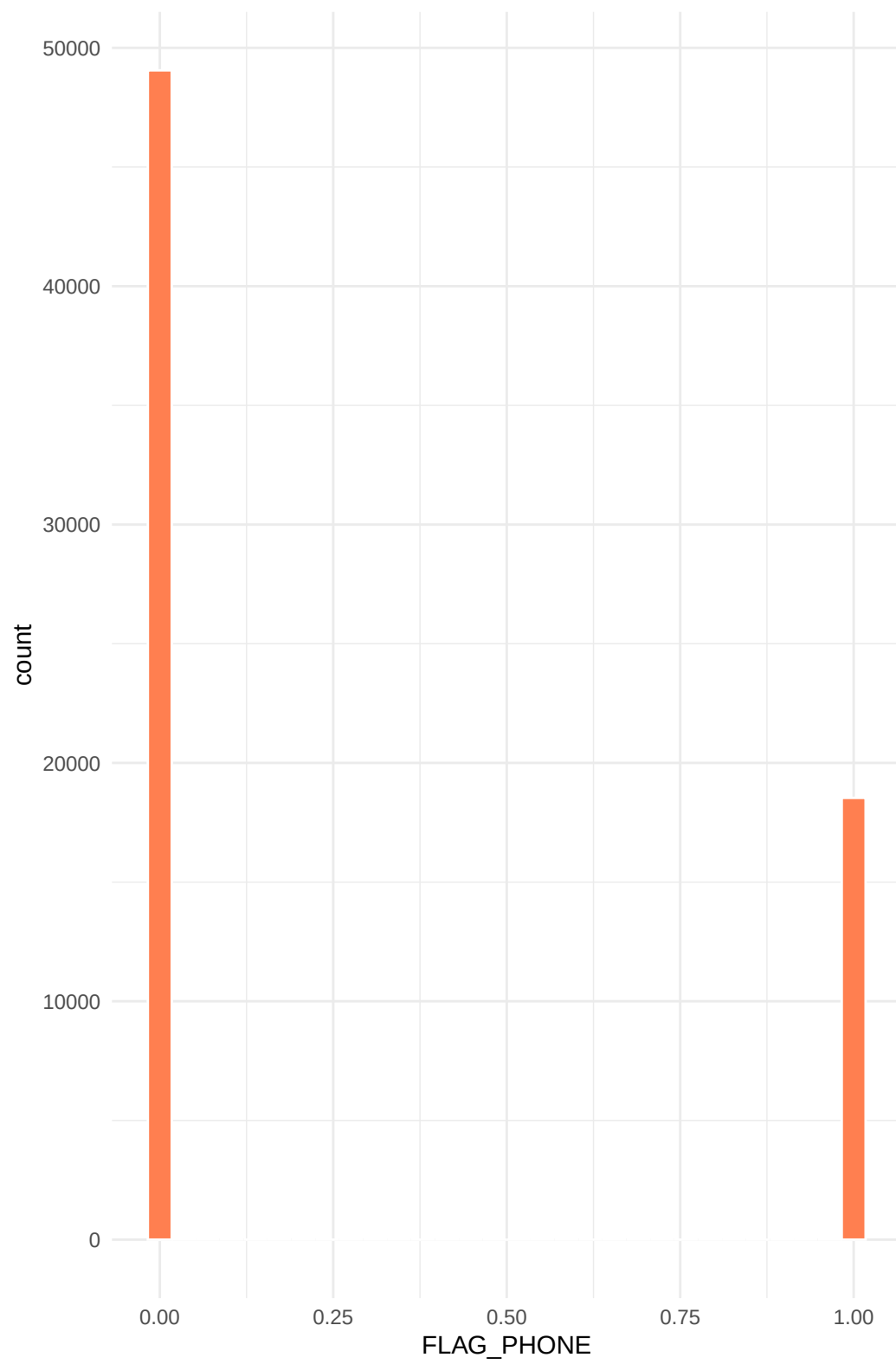
Distribution: FLAG_WORK_PHONE



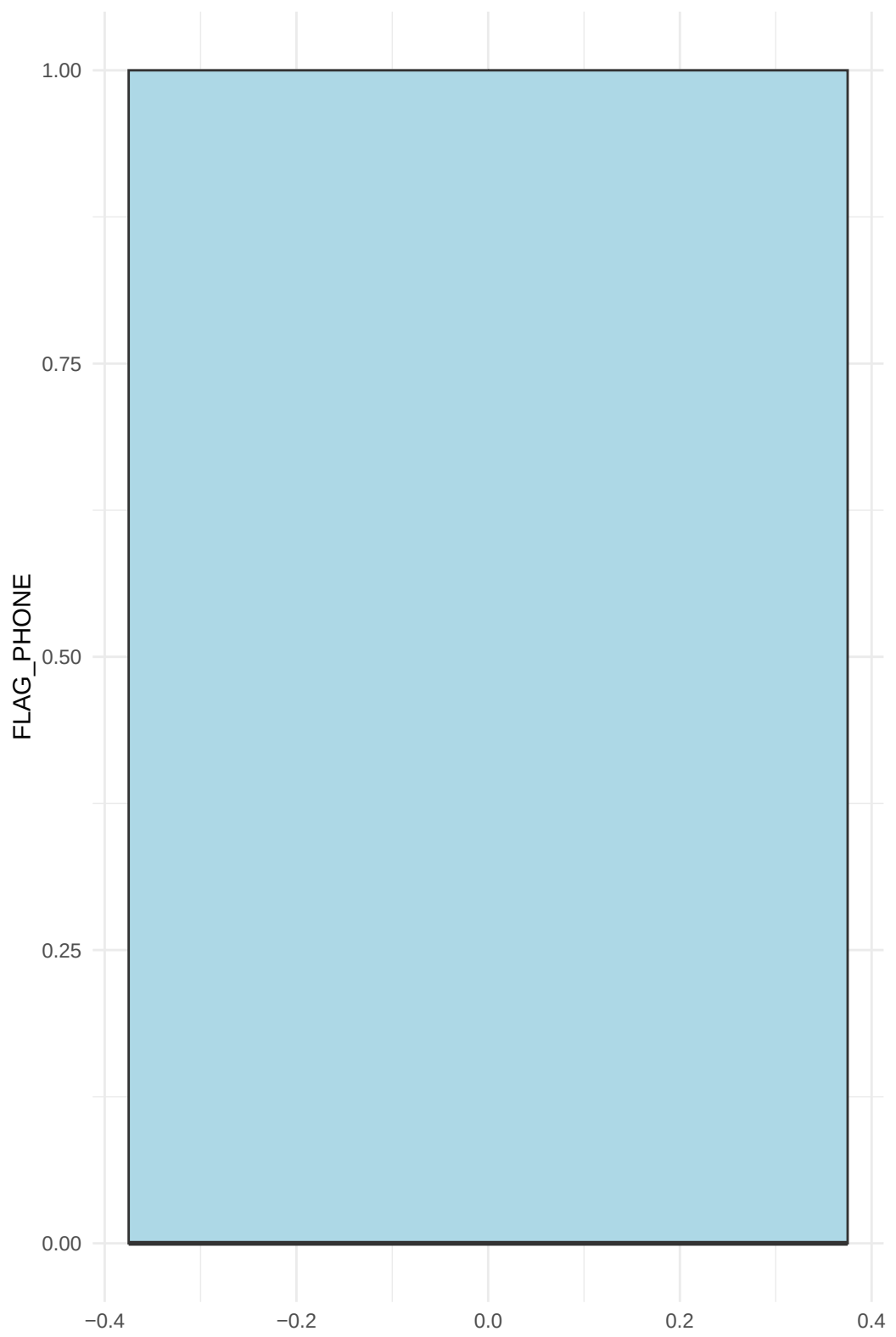
Outliers: FLAG_WORK_PHONE



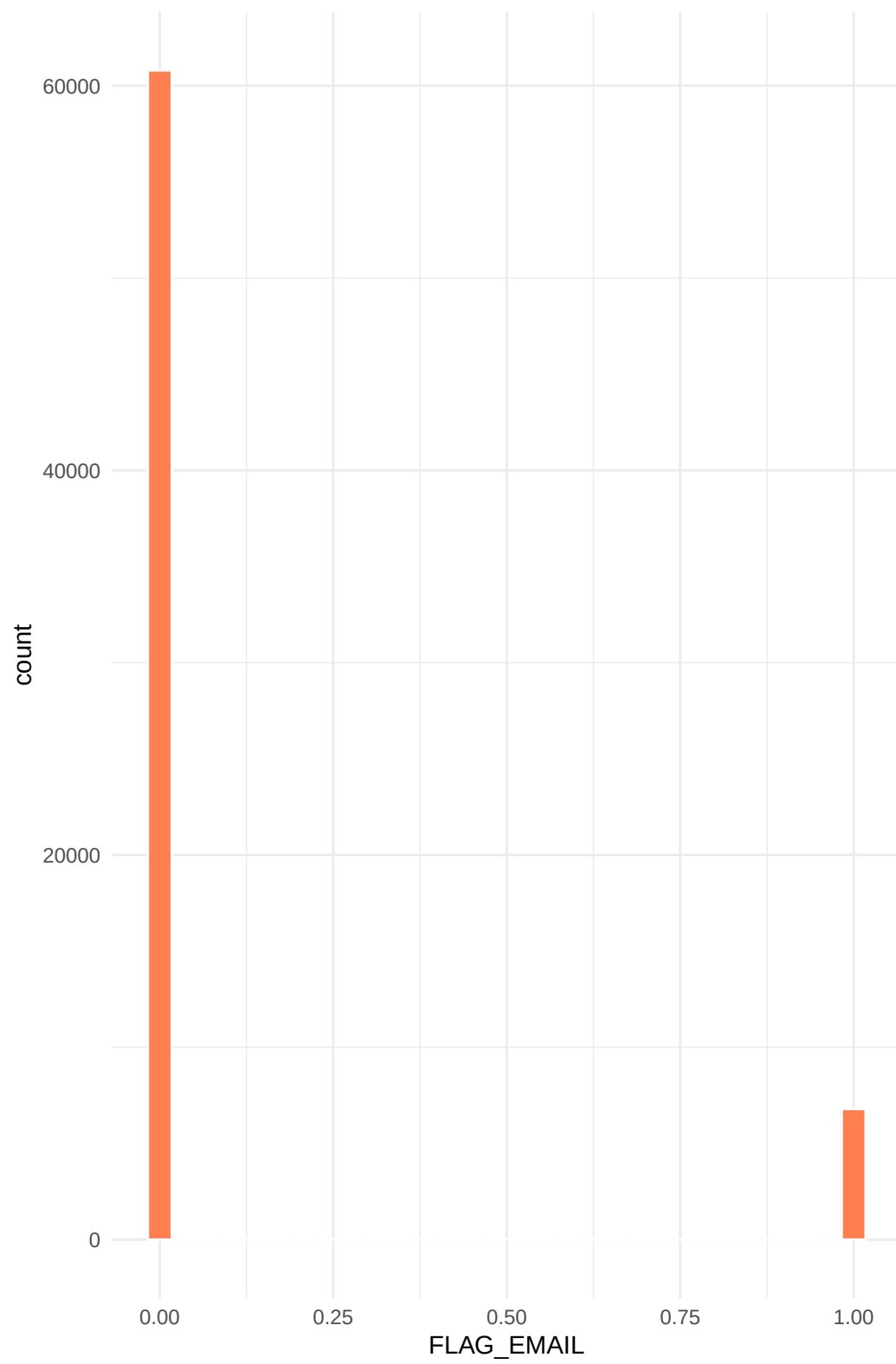
Distribution: FLAG_PHONE



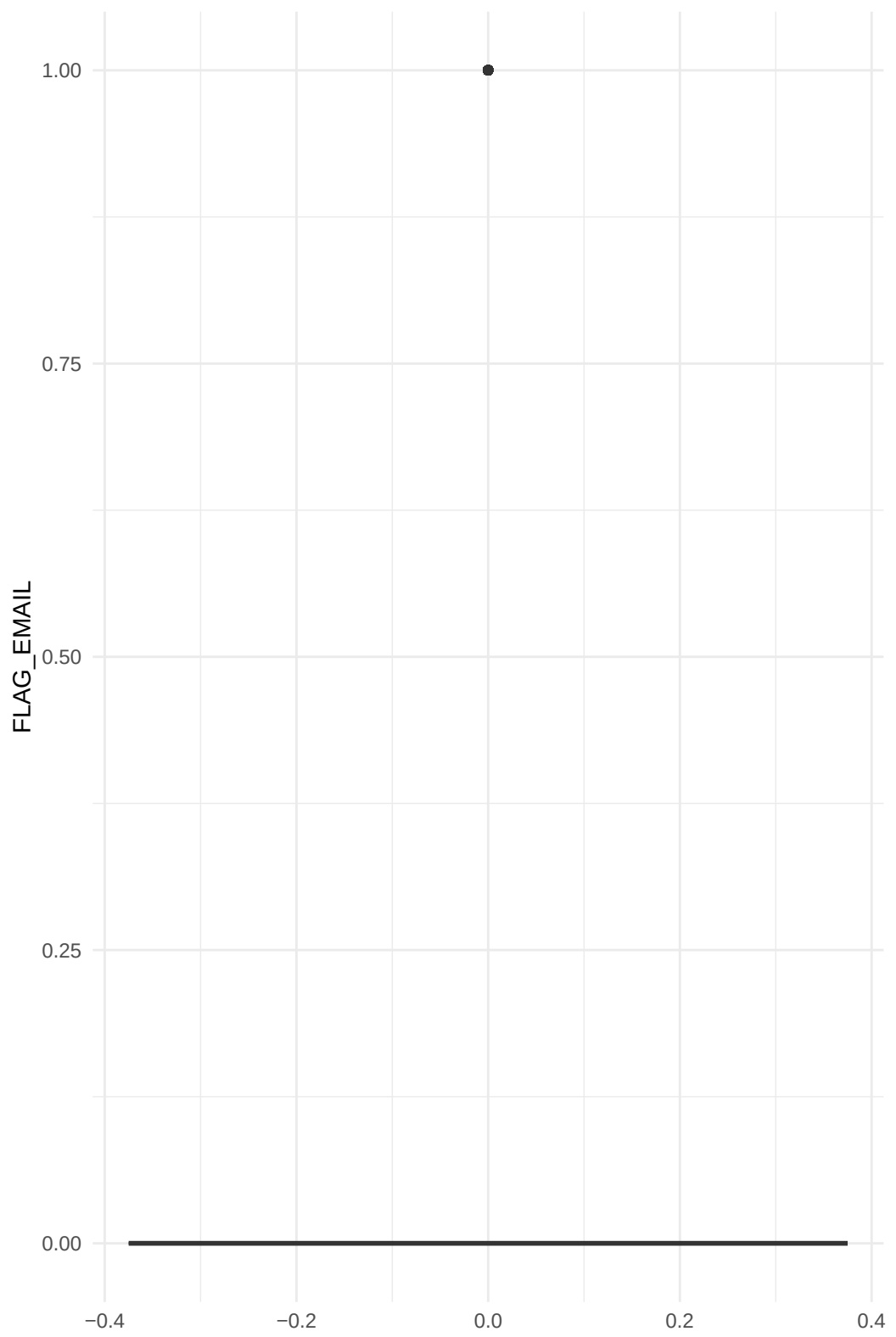
Outliers: FLAG_PHONE



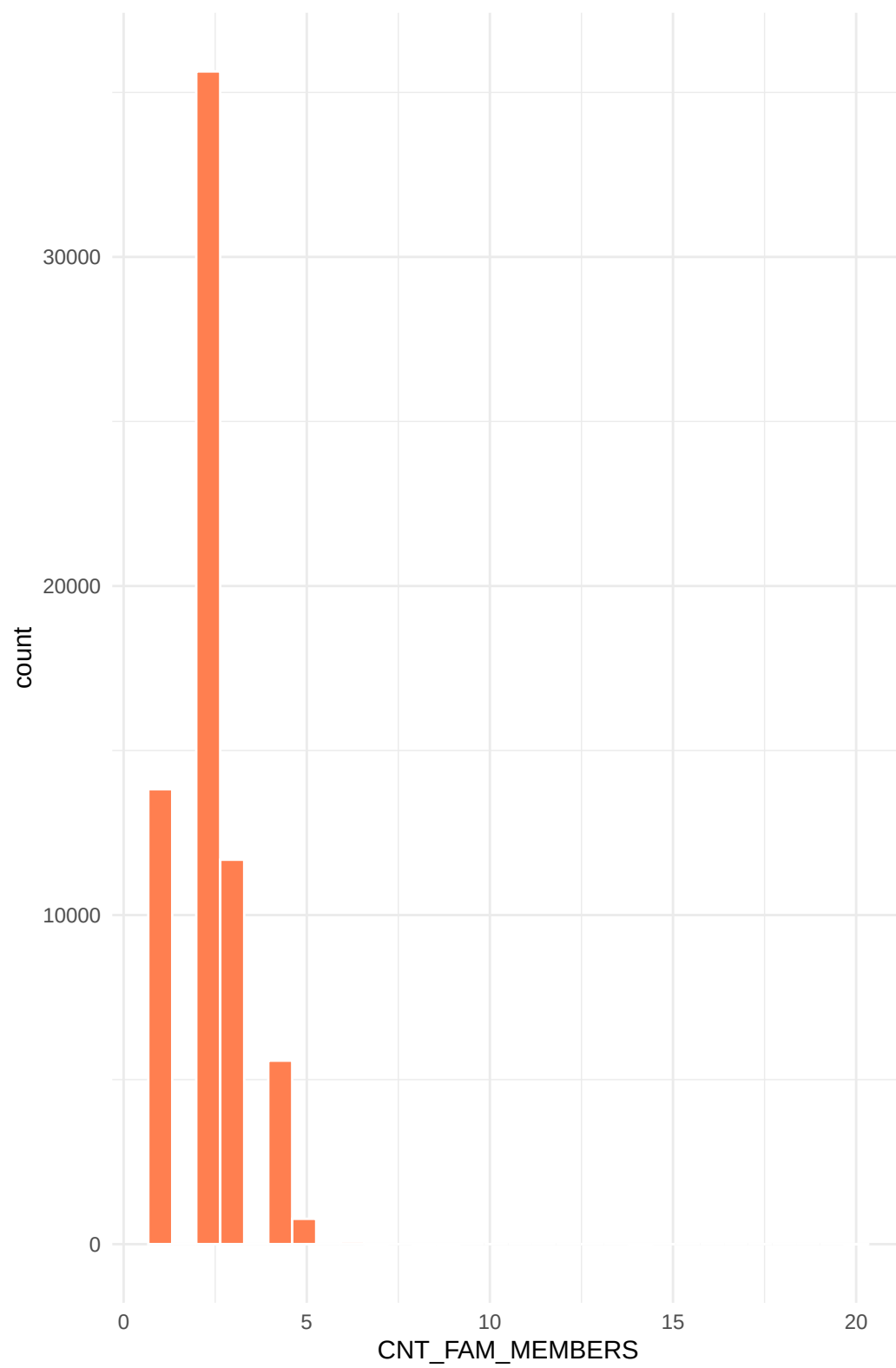
Distribution: FLAG_EMAIL



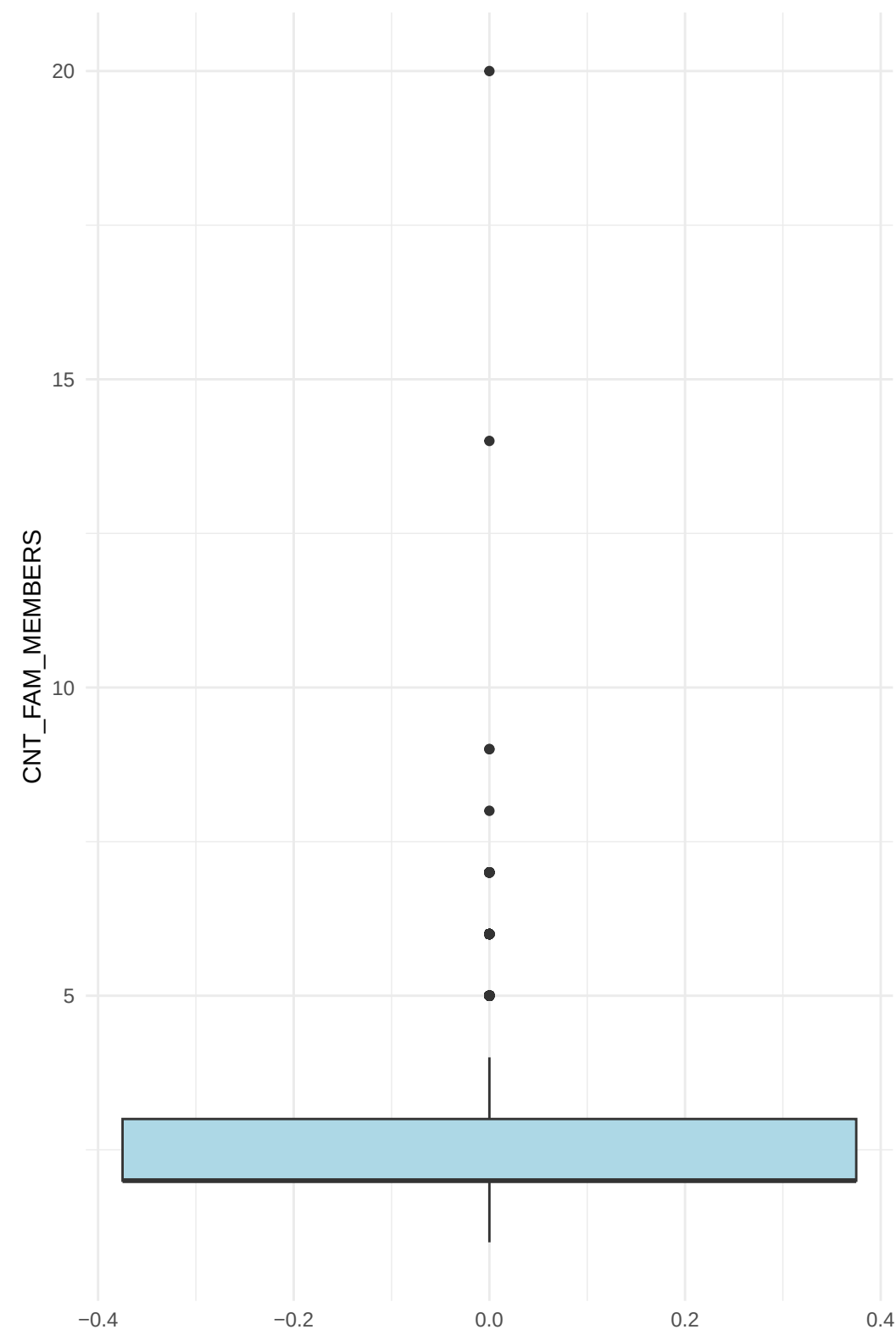
Outliers: FLAG_EMAIL



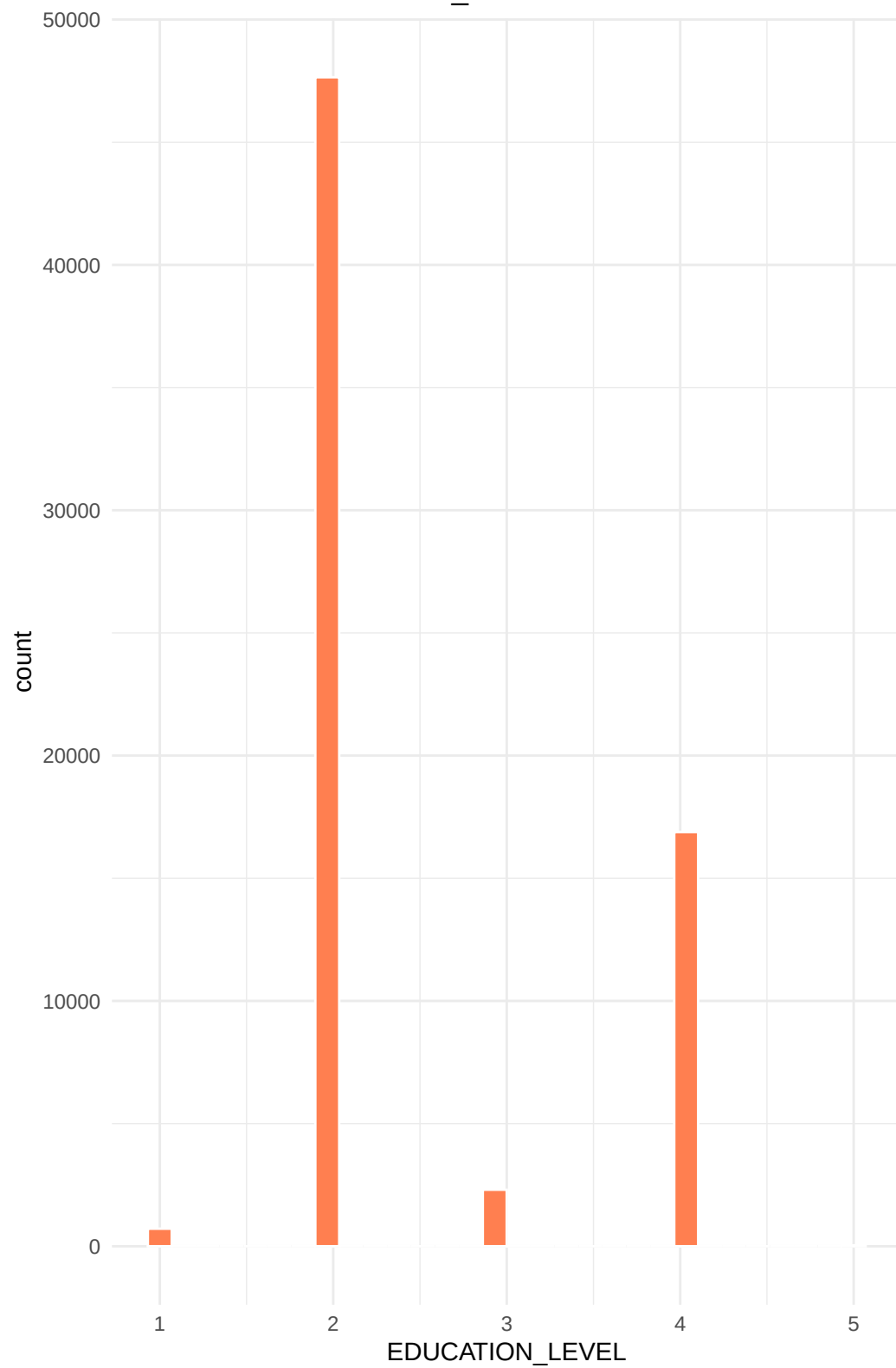
Distribution: CNT_FAM_MEMBERS



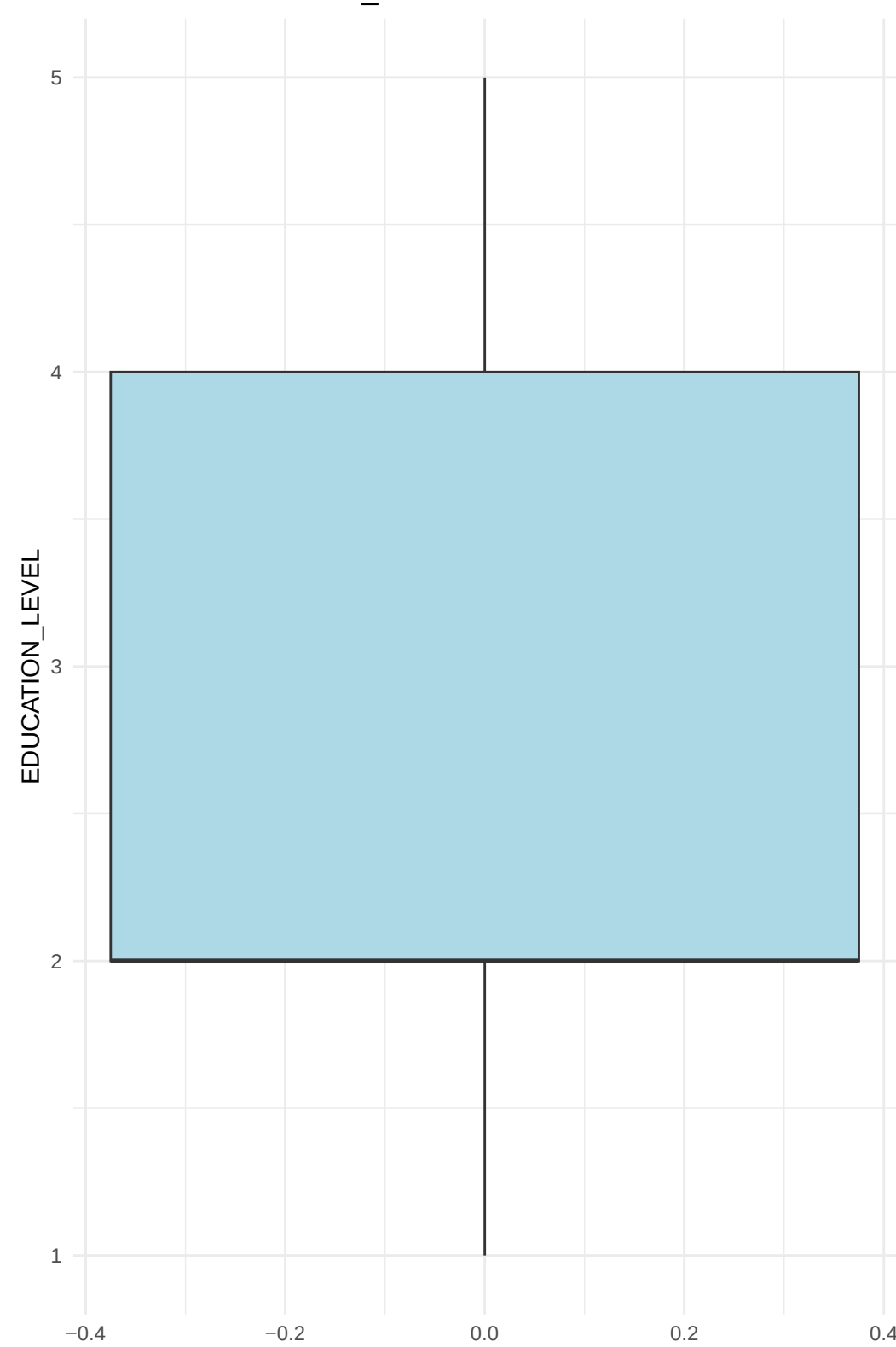
Outliers: CNT_FAM_MEMBERS



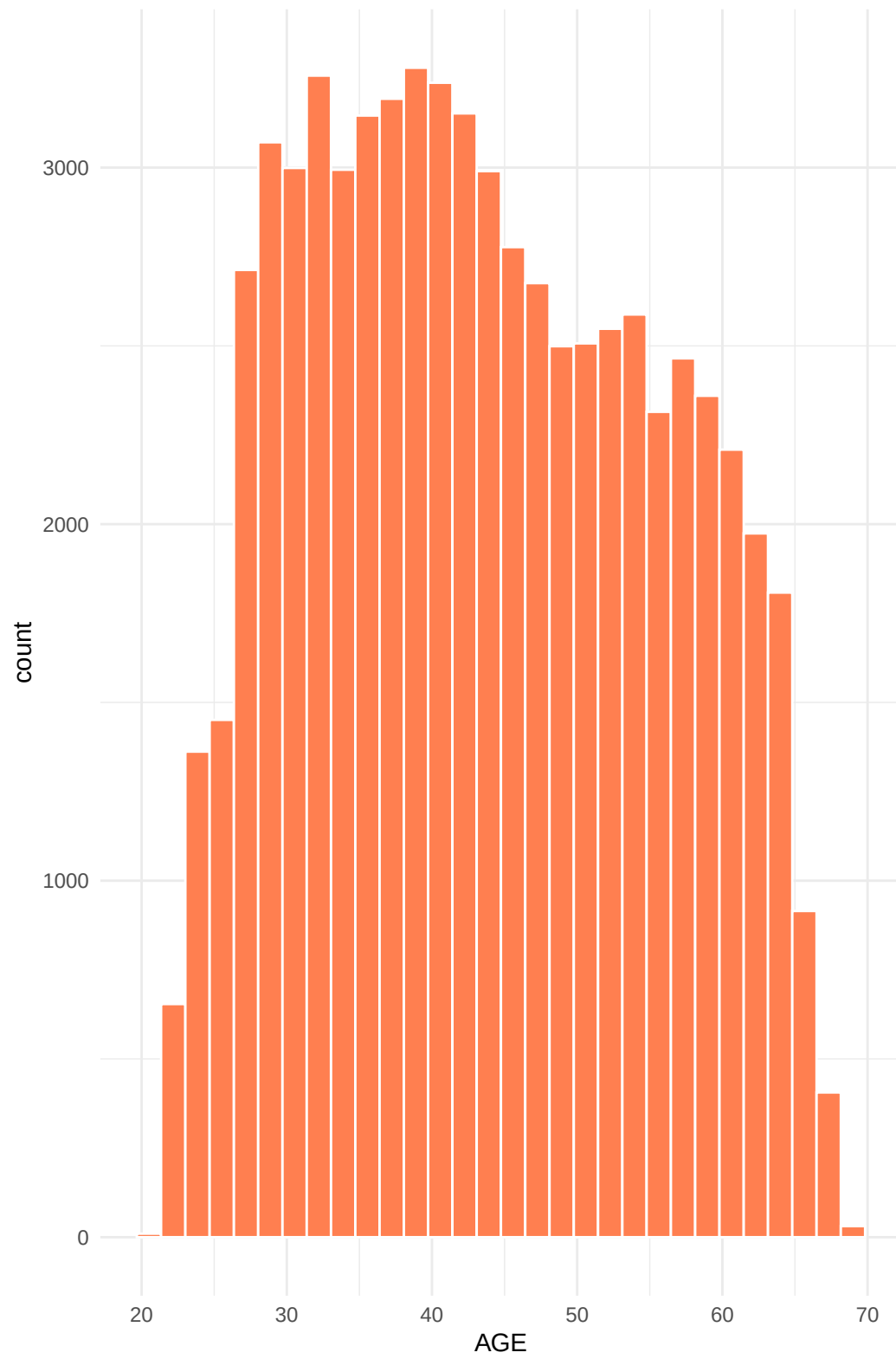
Distribution: EDUCATION_LEVEL



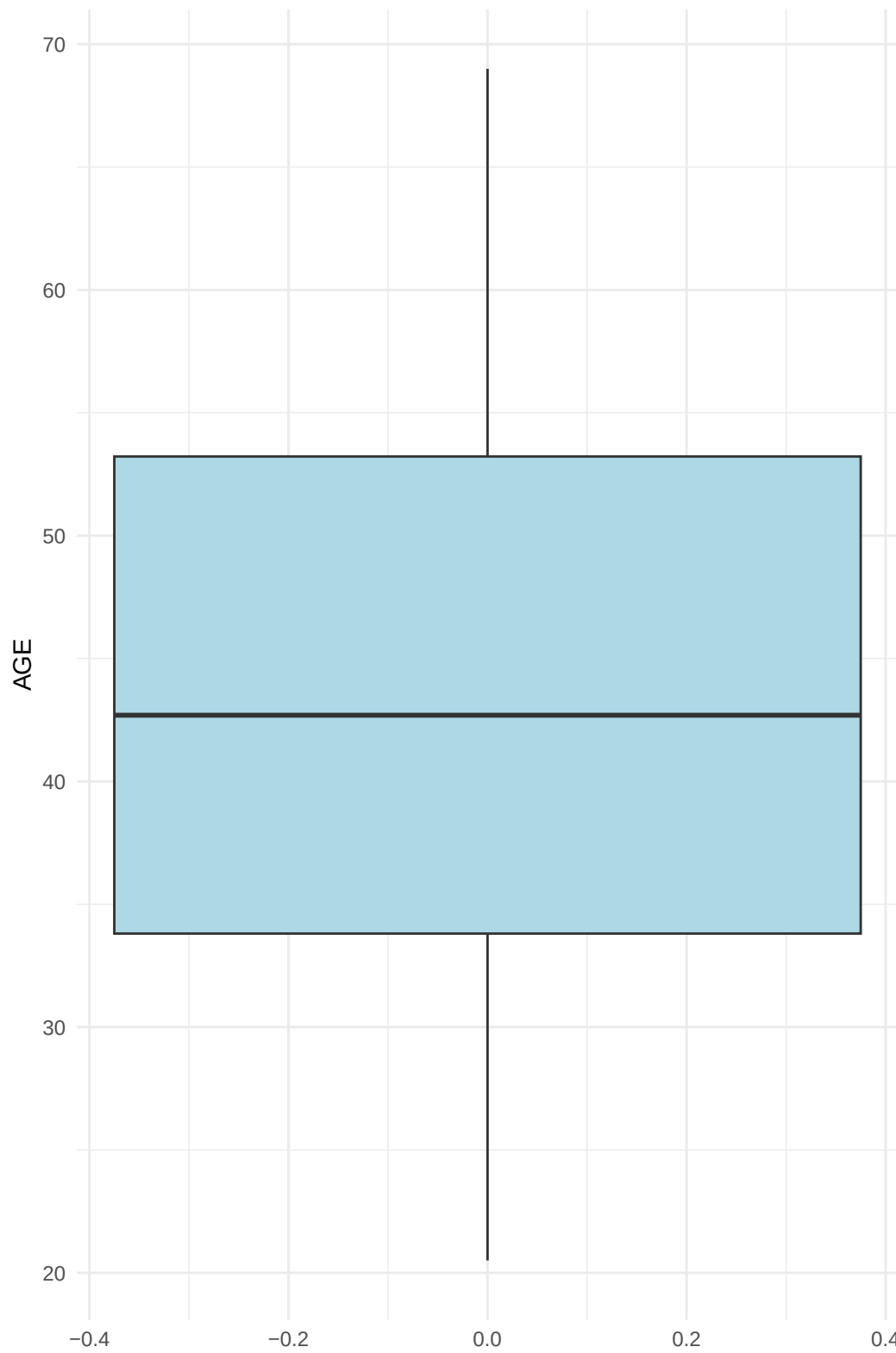
Outliers: EDUCATION_LEVEL



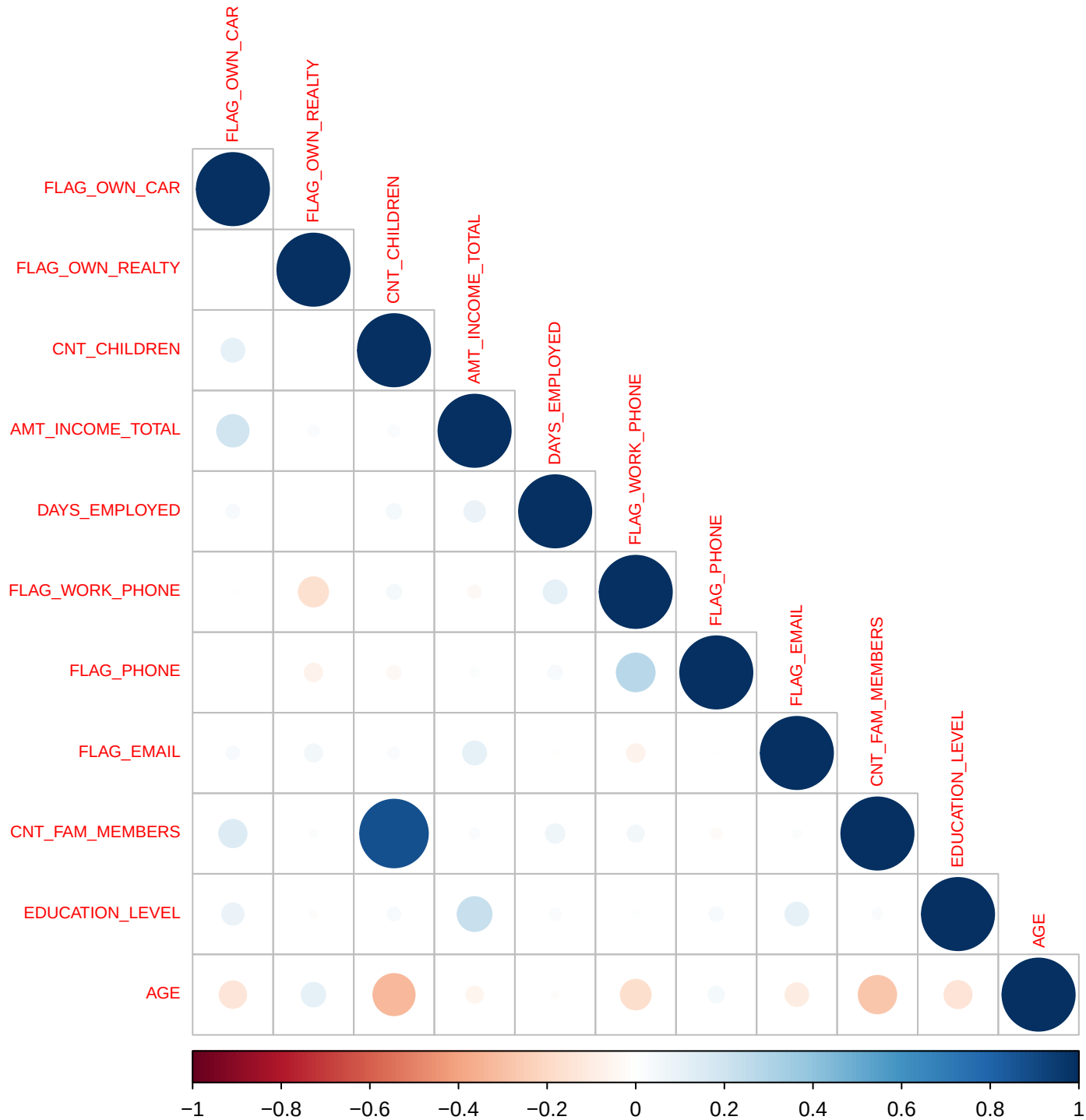
Distribution: AGE



Outliers: AGE



Correlation Matrix (Cleaned)



Structural Missingness

Occupation Status

Present

Prop

1

Commercial associate

Pensioner

State servant

Student

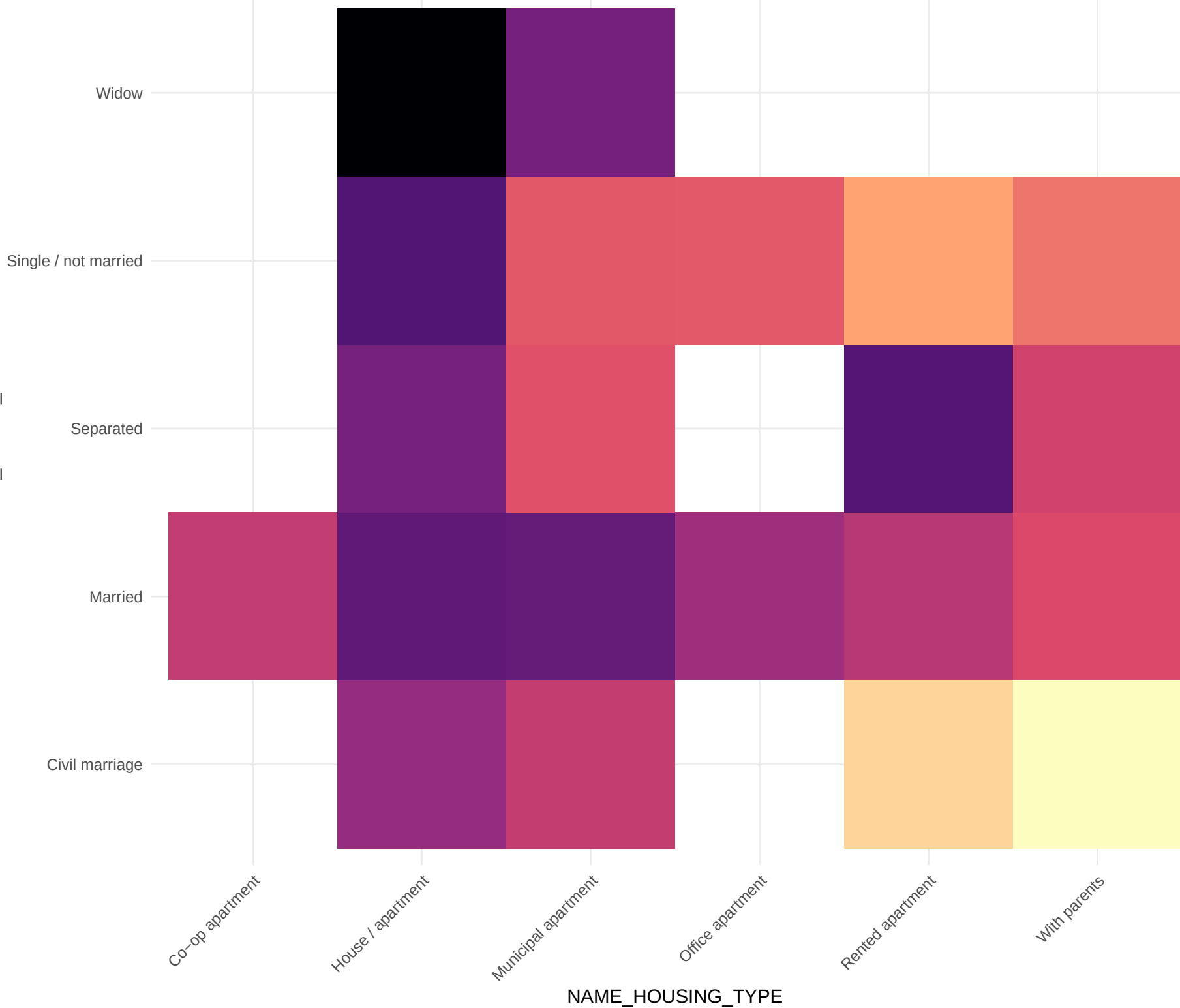
Working

Income Type



Risk Interaction: Family Status vs Housing

NAME_FAMILY_STATUS



Default Rate



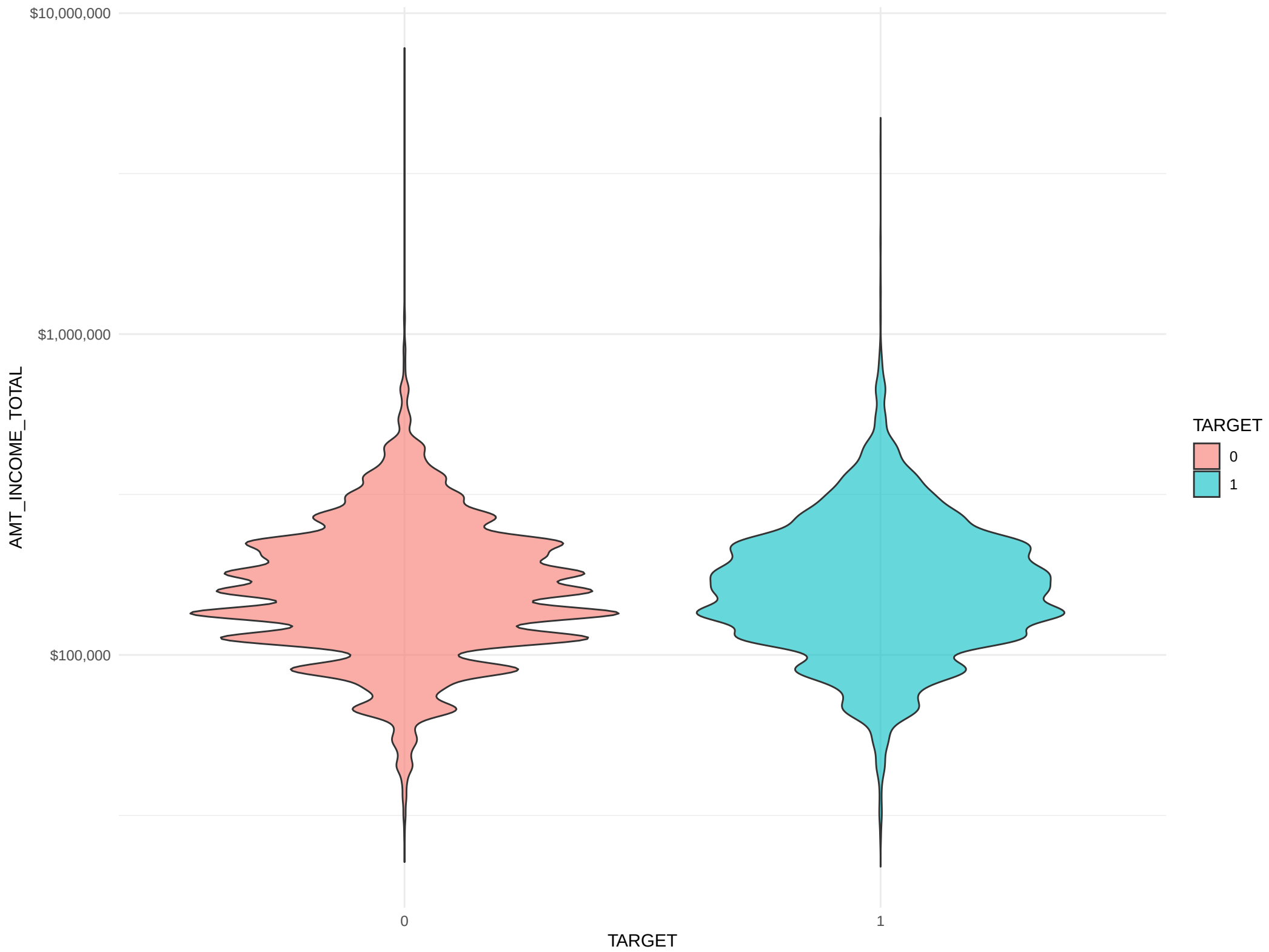
20%

15%

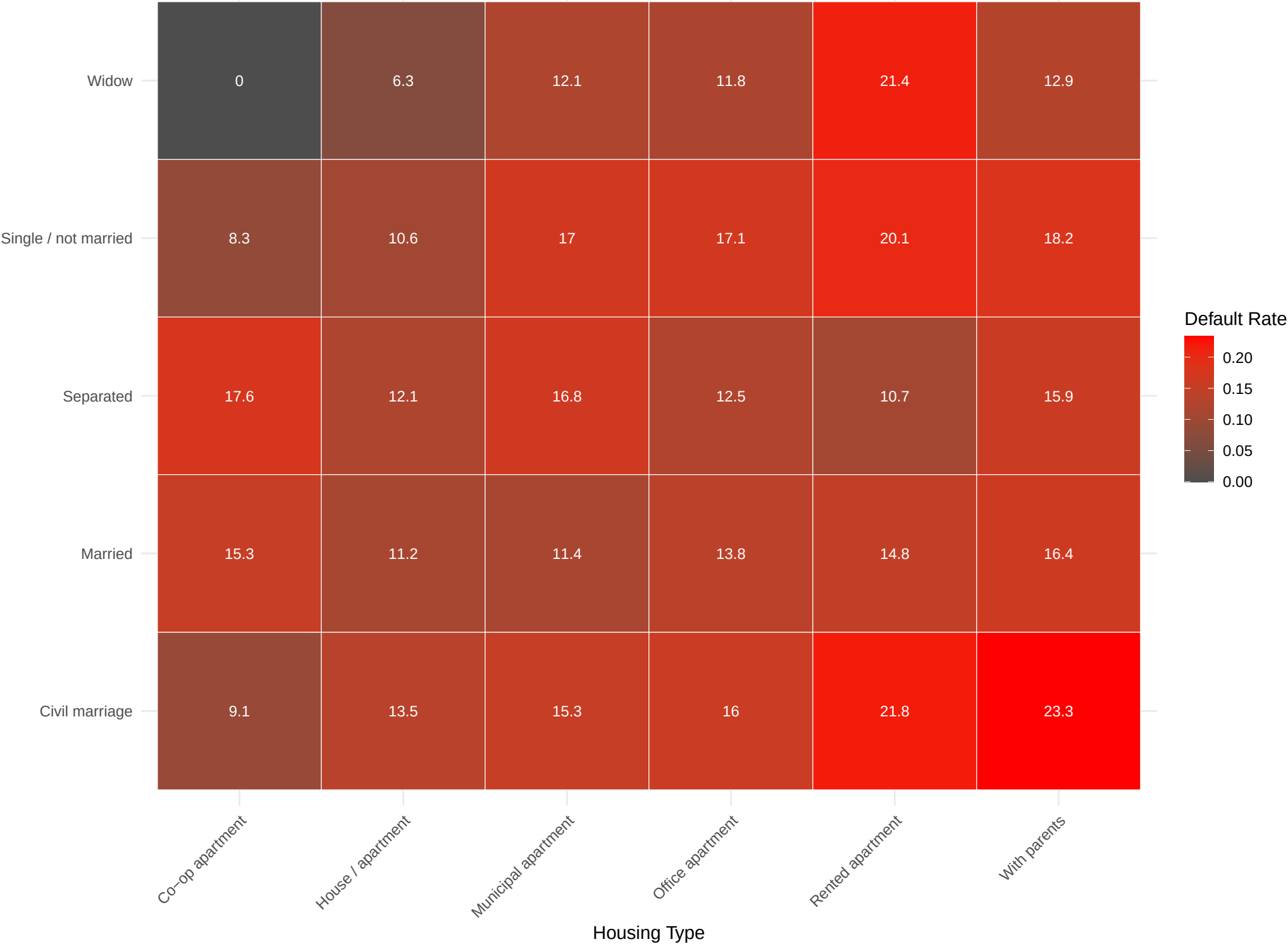
10%

NAME_HOUSING_TYPE

Income Distribution by Target Class (Log Scale)



Risk Heatmap: Family Status vs. Housing Type
Numbers represent Default Rate (%)



Risk Signal: Default Rate by Housing Type

reorder(NAME_HOUSING_TYPE, Default_Rate)

With parents

Rented apartment

Office apartment

Co-op apartment

Municipal apartment

House / apartment

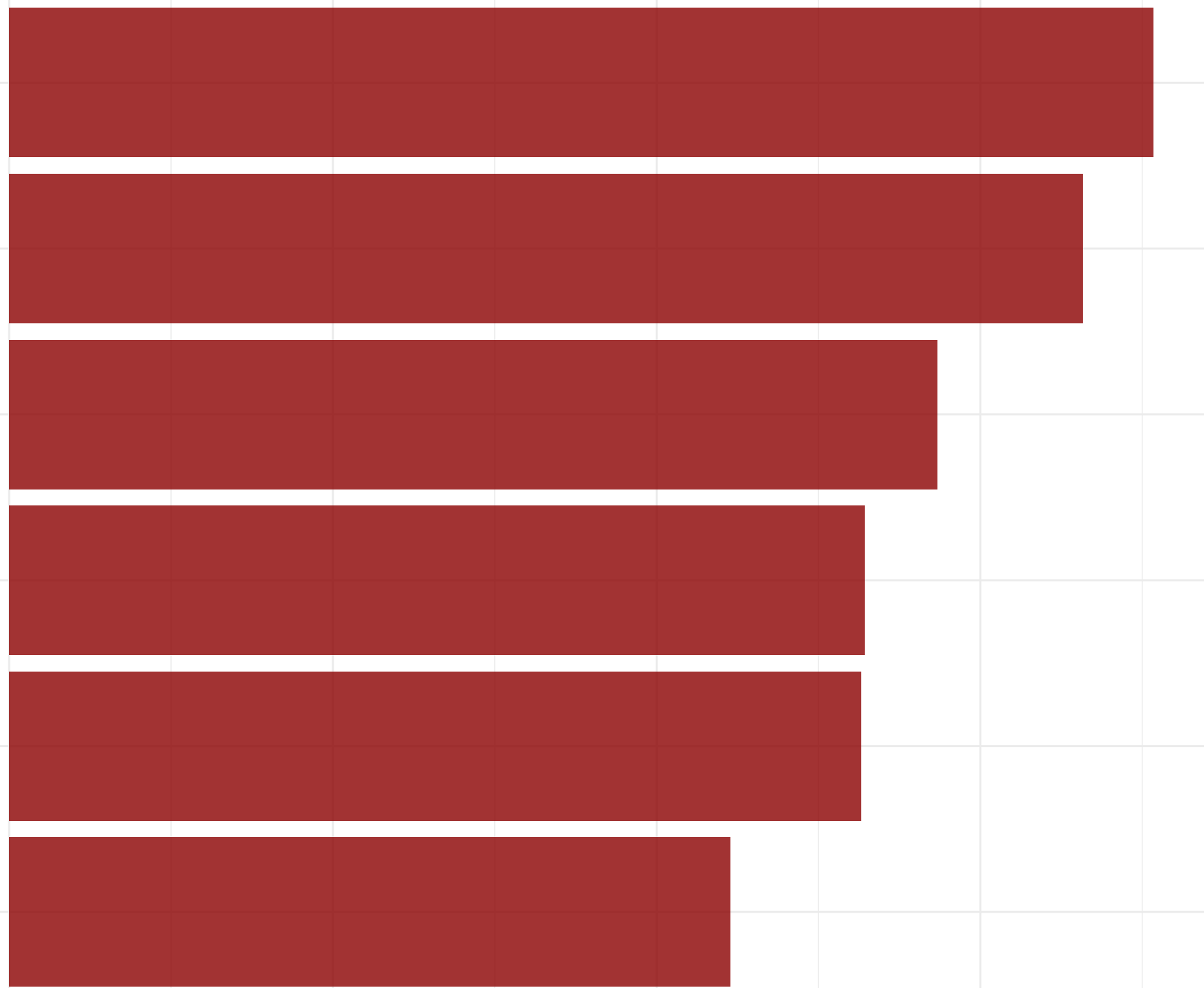
0.00

0.05

0.10

0.15

Default_Rate



Validation of Employment Flag

